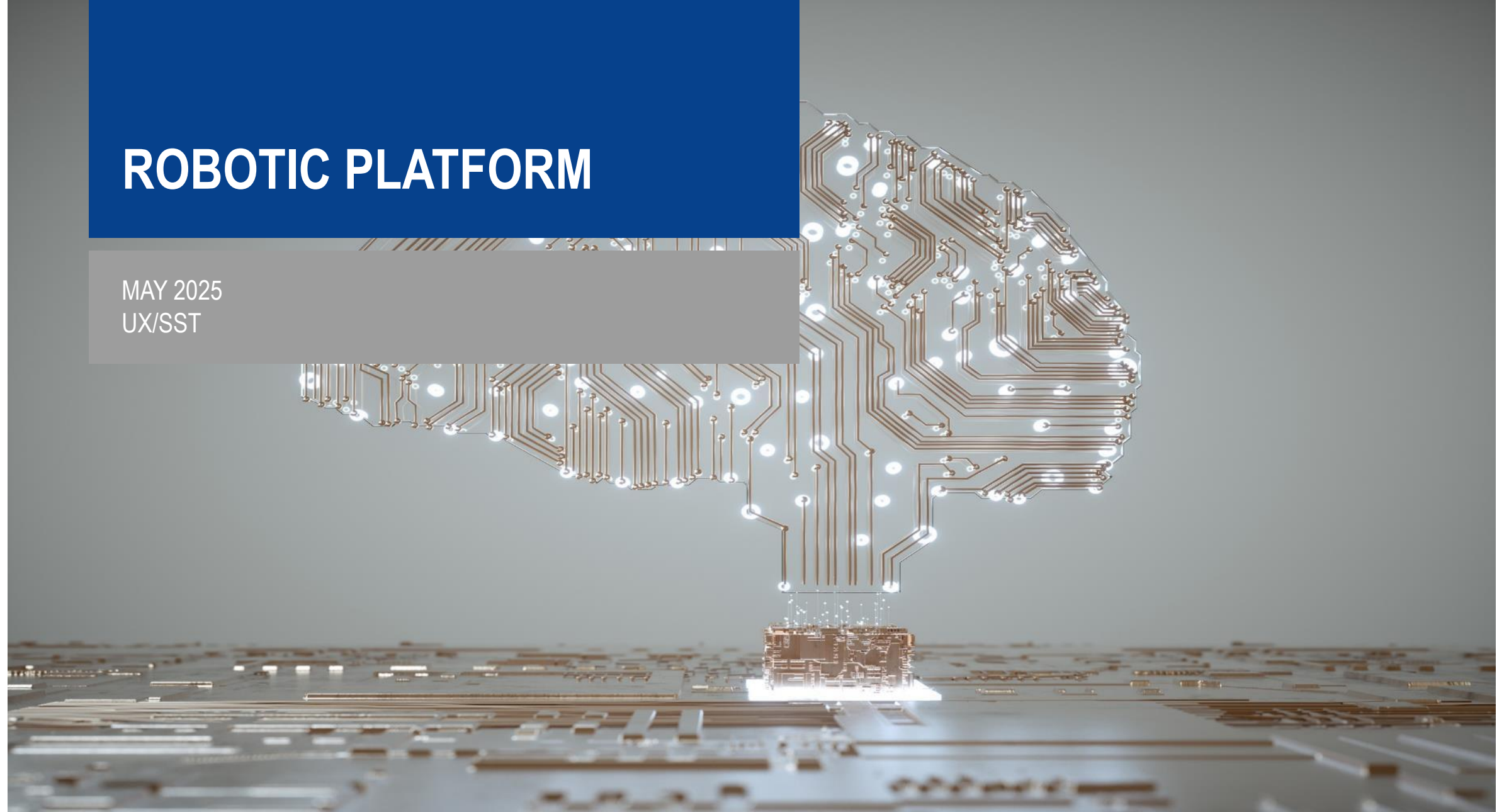


ROBOTIC PLATFORM

MAY 2025
UX/SST



AGENDA

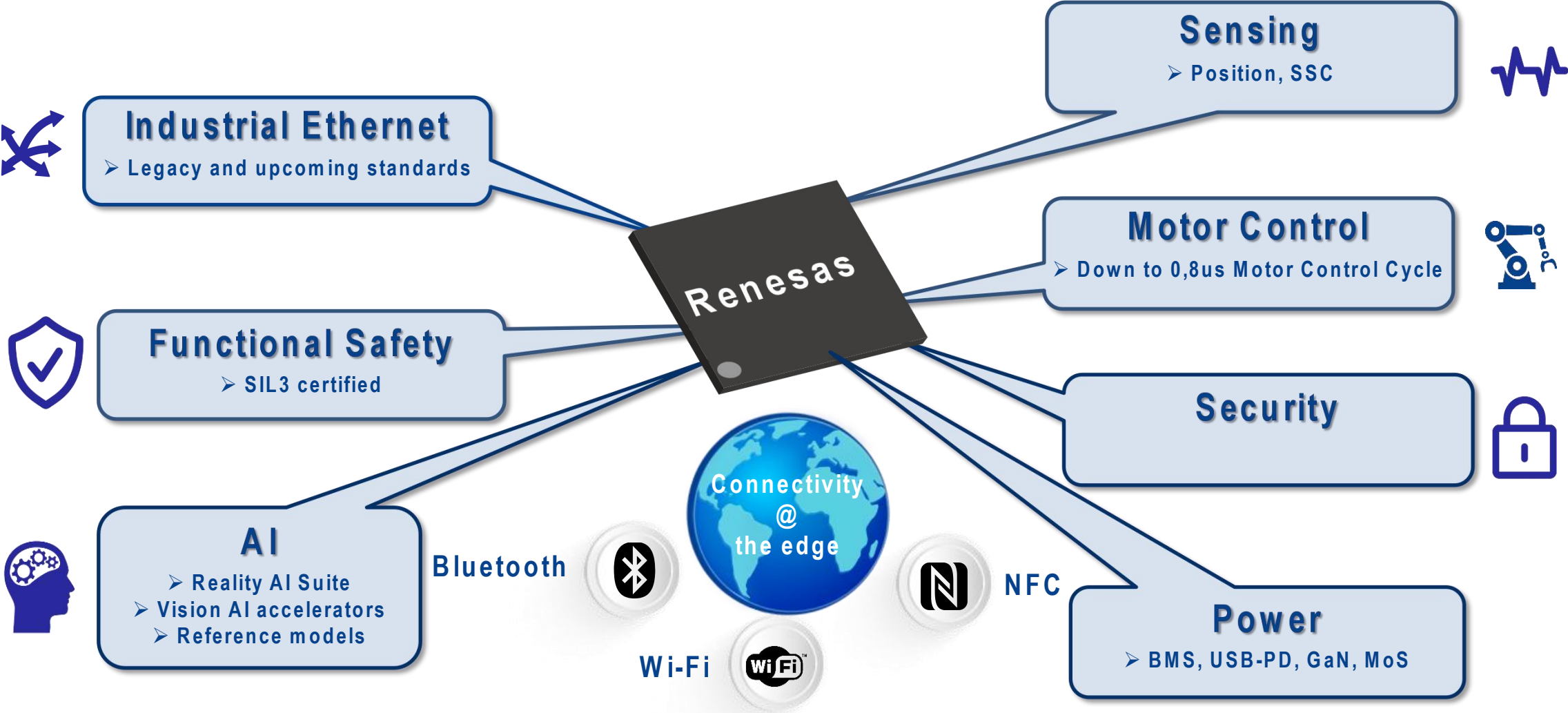


- Industrial automation market and platform overview:
- Renesas Solutions Building Blocks for Robotics:
 - IA
 - FUSA
 - AI and Humanoid Robotics

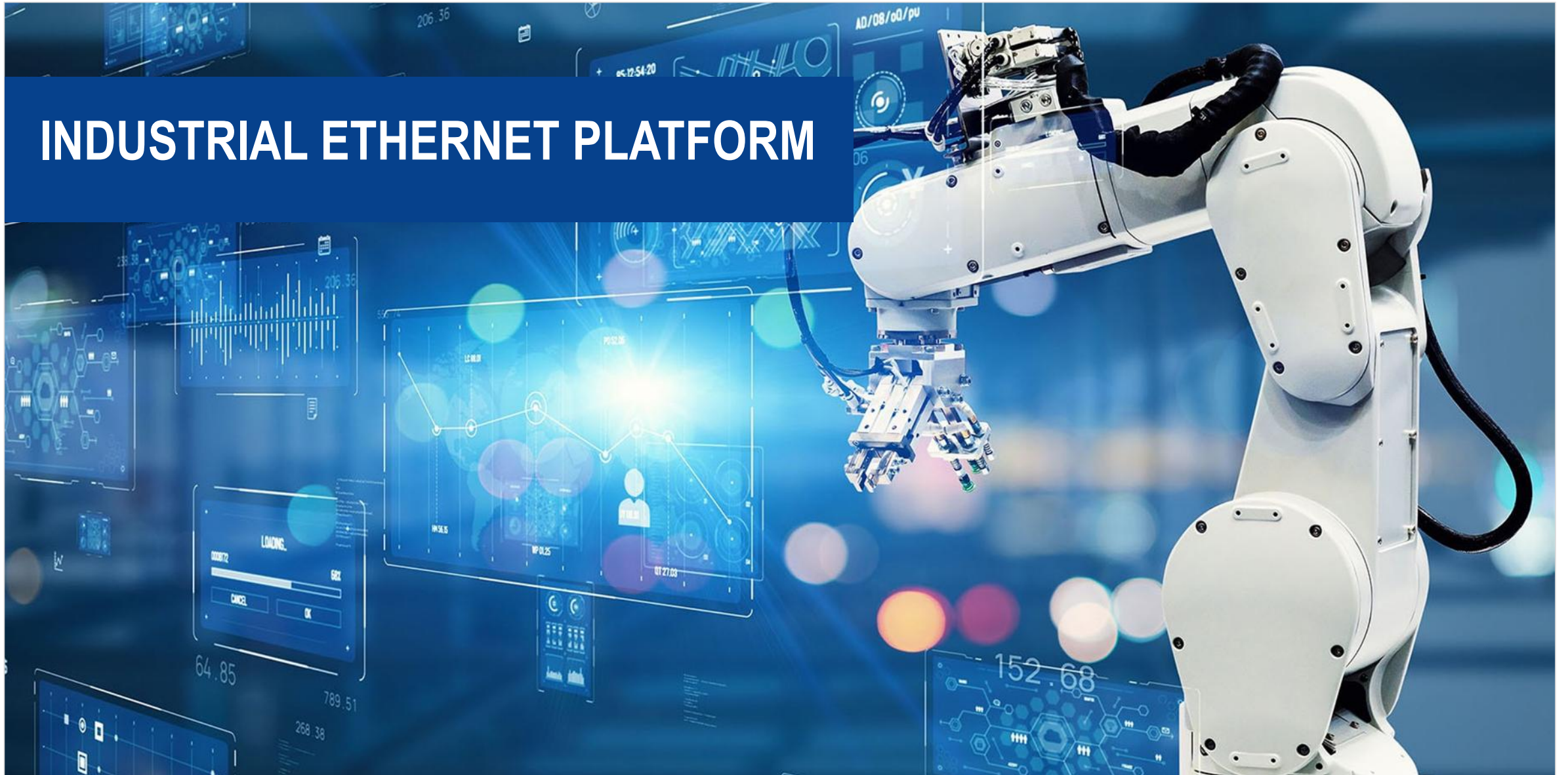
RENESAS PRODUCTS AND SOLUTIONS FOR ROBOTIC SEGMENT

Robotic core application	CAGR(25-30) 2024 TAM	Renesas Today Solutions	Solution Value and Win stories	Core devices for Solution	Market Trend and Renesas Solution in 2025
Industrial Robots 	- CAGR=7% 400k units/2024	- EtherCAT Servo/Gateway/Coupler/IO - Multi Industrial protocol	- We provide system level solution with multi protocols Win at all leading customers	- RZ/T,N - PMIC - Analog	- FUSA solution - Profinet + PROFIdrive Servo solution - Motion control + Multi joint control
Collaborative Robots (Cobots) 	- CAGR=30% 150k units/2024	- Joint control - Motor-encoder - HMI	High integration solution&product for Joint and encoder	- RZ/T, N - PMIC - Position sensor	- Motion control + Multi joint control - Vision AI solution
Mobile Robots 	- CAGR=15% 800k units/2024	- System control and motor control - HMI - Connectivity (BLE)	- System control and motor algorithm. - HMI solution with Linux and GUI.	- RA/RX(system control) - RZ/G(HMI) - BLE	- Vision AI solution - LLM - Connectivity(WiFi6+BLE)
Humanoid Robots 	- CAGR=75% 10k units/2024	- Dex-Hand - Joint control - Motor-encoder	Win at the leading customers for the Joints, and system control	- Position sensor - RZ/T, N - RZ/V2H - RCAR	- Renesas Robotic development kit and AI framework - High computing vision AI + Motion AI

RENESAS BUILDING BLOCKS FOR ROBOTICS



INDUSTRIAL ETHERNET PLATFORM

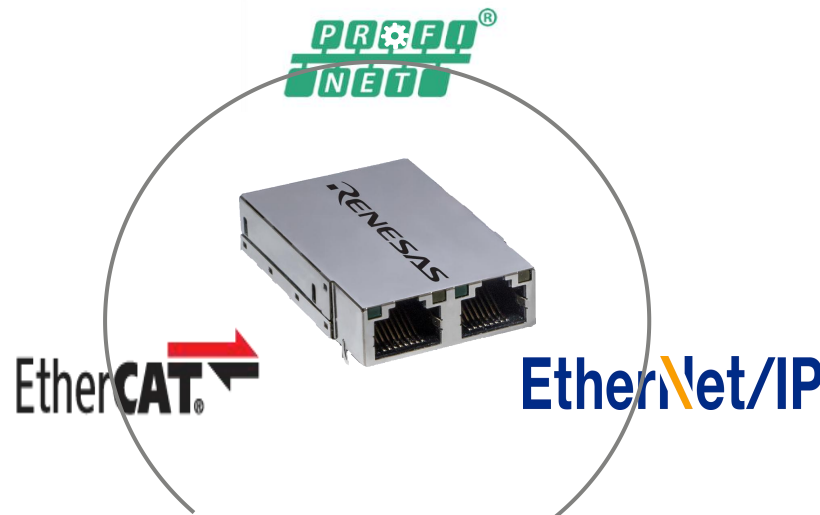
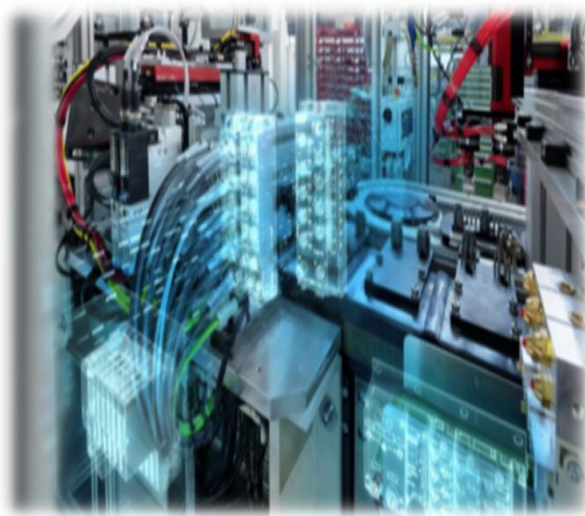


WHAT IS INDUSTRY ETHERNET

Industry Ethernet, an advanced technology is needed in industrial settings to ensure specific manufacturing data is correctly sent and received. If using an example of a bottle-filling plant, industrial Ethernet automation technology is able to send filling data over the network to ensure the bottles are being filled as intended

In order to achieve better determinism, industrial Ethernet uses specialized protocols in conjunction with Ethernet. The more popular industrial Ethernet protocols are PROFINET®, EtherNet/IP®, EtherCAT®, SERCOS III, and POWERLINK®.

With industrial Ethernet, data transmission rates range from 10 Mbps to 1 Gbps.⁵ However, 100 Mbps is the most popular speed used in industrial Ethernet applications.



TARGET APPLICATION

FA (Factory Automation)

Beyond FA



AC Servo



Inverter



Remote I/O
Network module



Industrial Robot



Semiconductor
manufacturing equipment



CNC



Robot gripper



Conveying device



Knitting machine



Textile machine



Collaborative Robots



Production Printer



sewing machine



Wind turbine



Elevator



Medical equipment



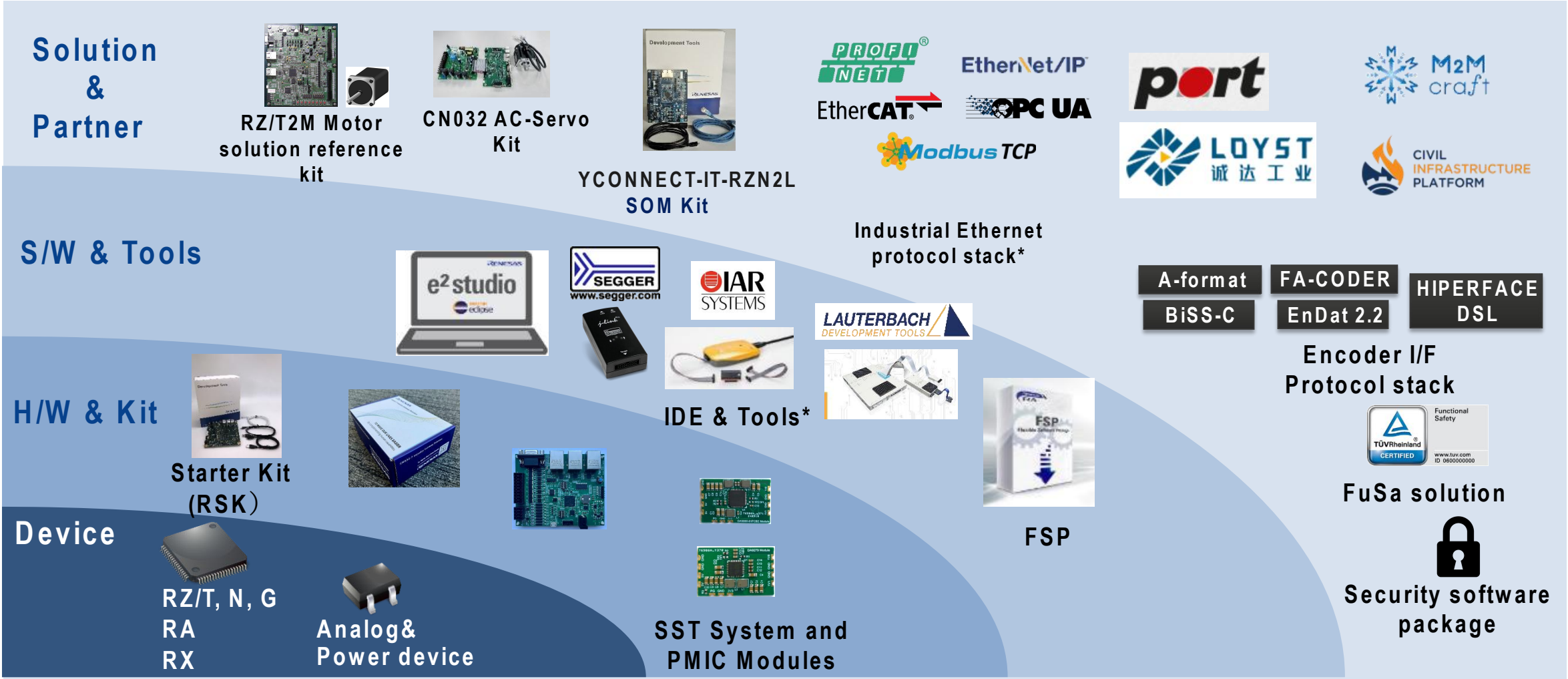
Charging station



railroad

ECOSYSTEM OVERVIEW

DEVELOPMENT ENVIRONMENT / SOLUTION TO SUPPORT CUSTOMER PRODUCT DEVELOPMENT



RENESAS RZ FAMILY

HIGH-PERFORMANCE APPLICATION FOCUSED AND GENERAL-PURPOSE PROCESSORS

Vision AI

RZ/V Series

AI accelerator + Linux

Industrial Network

RZ/N Series

Multi-protocol Industrial Ethernet with Redundancy + Linux/RTOS

Industrial real-time control

RZ/T Series

Multi-protocol Industrial Ethernet Real-time control + Linux/RTOS



Human Machine Interface & IoT Edge

RZ/G Series

3D Graphics + Linux/Android™

RZ/A Series

2D Graphics + RTOS

Vision AI

Renesas own cutting-edge AI technology "DRP-AI" for embedded AI application

Network

TSN and Multi-protocol Industrial network support

Real-time

High-performance Processing and High Precision Realtime Control

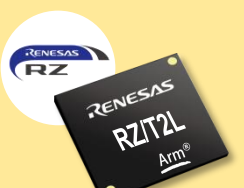
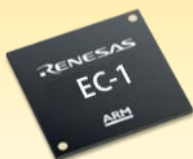
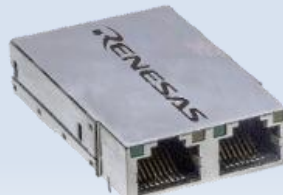
HMI

High-performance CPU operation and Graphics/Video Processing

Edge IoT

High-performance CPU and High-speed interfaces for IoT Edge products

PRODUCT LINEUP FOR INDUSTRIAL ETHERNET

Multi Protocol						EtherCAT	PROFINET
							
RZ/T1 Cortex®-R4 + Cortex®-M3 CPUs* 2 port w/ Switch EtherCAT Slave Controller	RZ/T2M Cortex®-R52 x2 Trigonometric Acc. 3port GMAC w/ switch TSN Switch EtherCAT Slave	RZ/N1 Dual Cortex®-A7 + Cortex®-M3 CPU 5port GMAC w/ switch True IE multi-protocol PRP/HSR/MRPD	RZ/N2L Cortex®-R52 Gigabit EtherMAC 3port GMAC w/ switch TSN Switch EtherCAT Slave	R-IN32M4-CL3 Cortex®-M4 CPU 2Port w/ switch gigabit PHY CC-Link IE Field and CC-Link IE TSN Class B	R-IN32M3-EC Cortex®-M3 CPU 2port switch w/ PHY EtherCAT Slave Controller	RZ/T2L Cortex®-R52 Trigonometric Acc EtherCAT Slave Controller	TPS-1 2Port with PHY PROFINET CPU <i>PROFINET IRT</i>
   	TSN    <i>PROFINET IRT</i>  	     	TSN    <i>PROFINET IRT</i>  	CC-Link IE TSN    	   	 EC-1 Cortex®-R4 EtherCAT Slave Controller	Multi Protocol  R-IN32M3 Module RJ-45 x2 SPI for Application CPU

ROBOTIC SERVO CONTROL SOLUTION ROADMAP

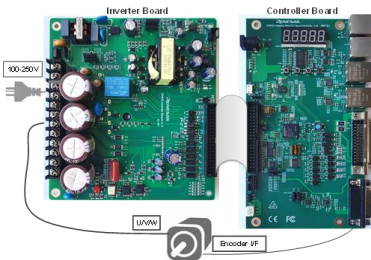
COVERED WILDER ENCODER MOTOR AND INDUSTRIAL ETHERNET USE CASES IN ROBOTIC

Servo and Inverter

RZ/T2M

RZ/ N2L

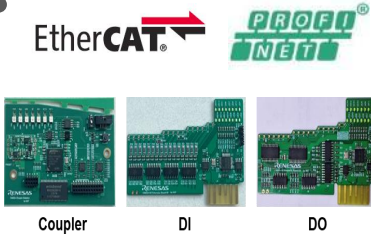
RZ/T2L



Gateway and Remote I/O

Gateway

Remote I/O



Profinet IRT

CN032-7 T2M



RZ/T2H
Motion
Control

2021-2023

May 24

July 24

Jan 25

2025

Industrial motor drives

RZ/T2M

RZ/T2L



AC drive / Inverter
w/ Ind. Network, FuSa Ind. Network, FuSa

AC servo w/
Ind. Network, FuSa

Network controllers

RZ/N2L



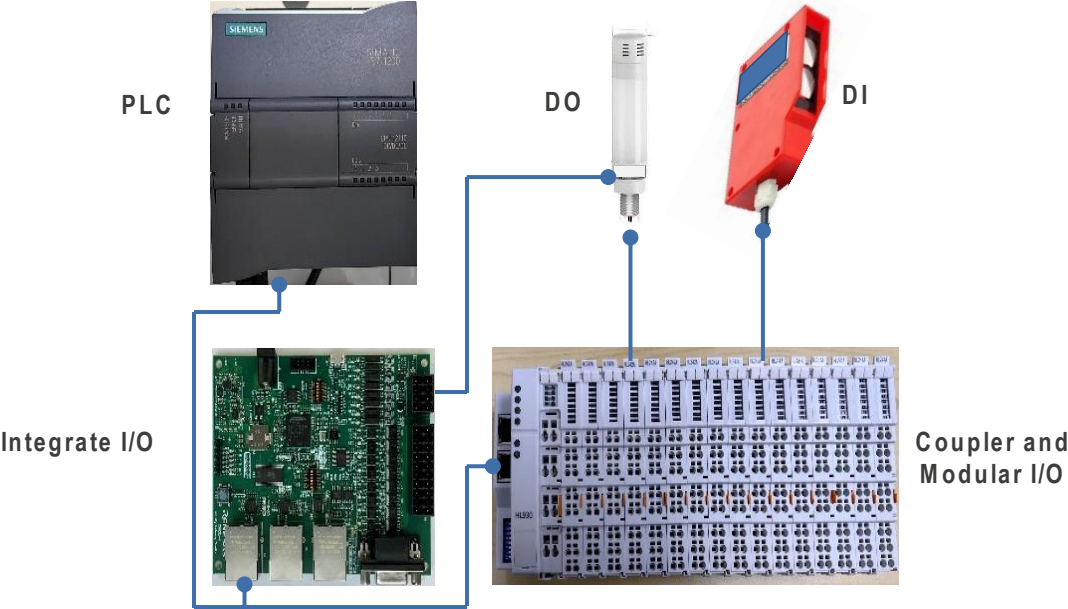
Remote I/O,
Gateway for IA, BA, PA..
Sensor hub



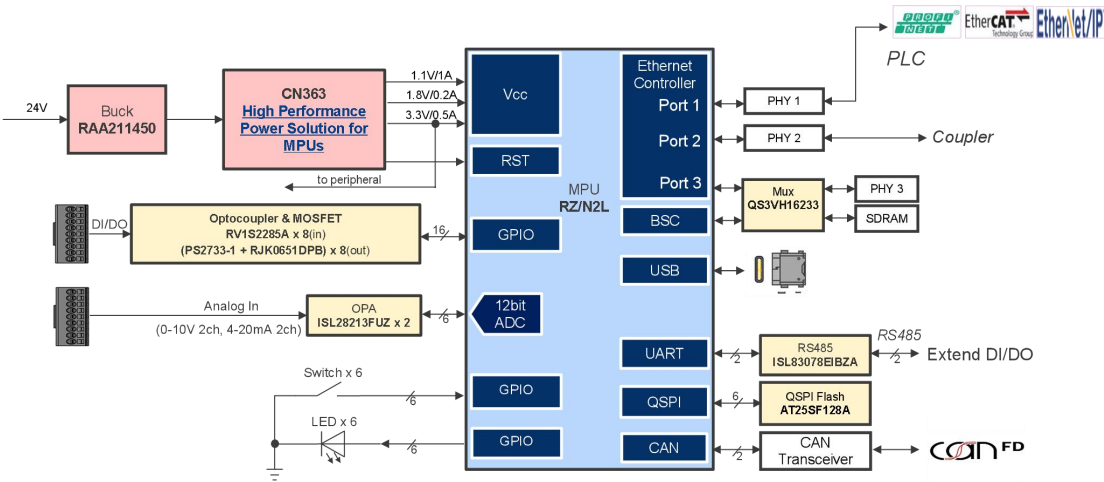
Industrial sensor
w/ Industrial Network

INDUSTRIAL GATEWAY/COUPLER/REMOTE IO SYSTEM BLOCK

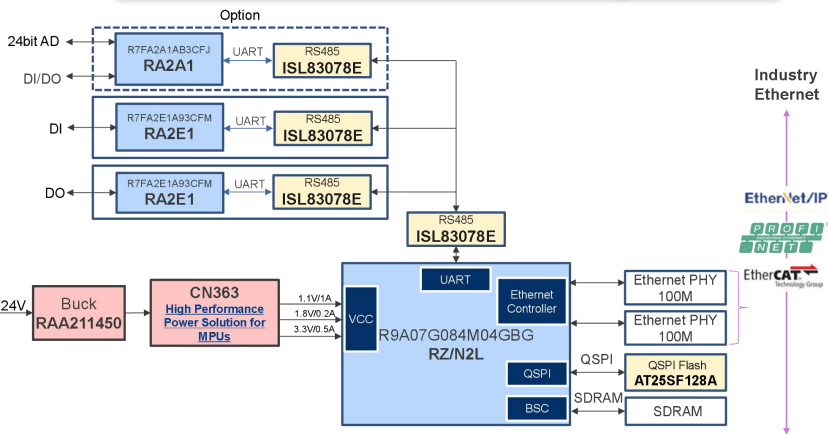
Industry Gateway/Coupler/Remote IO System



Gateway/Coupler Block Diagram



Remote I/O Block Diagram



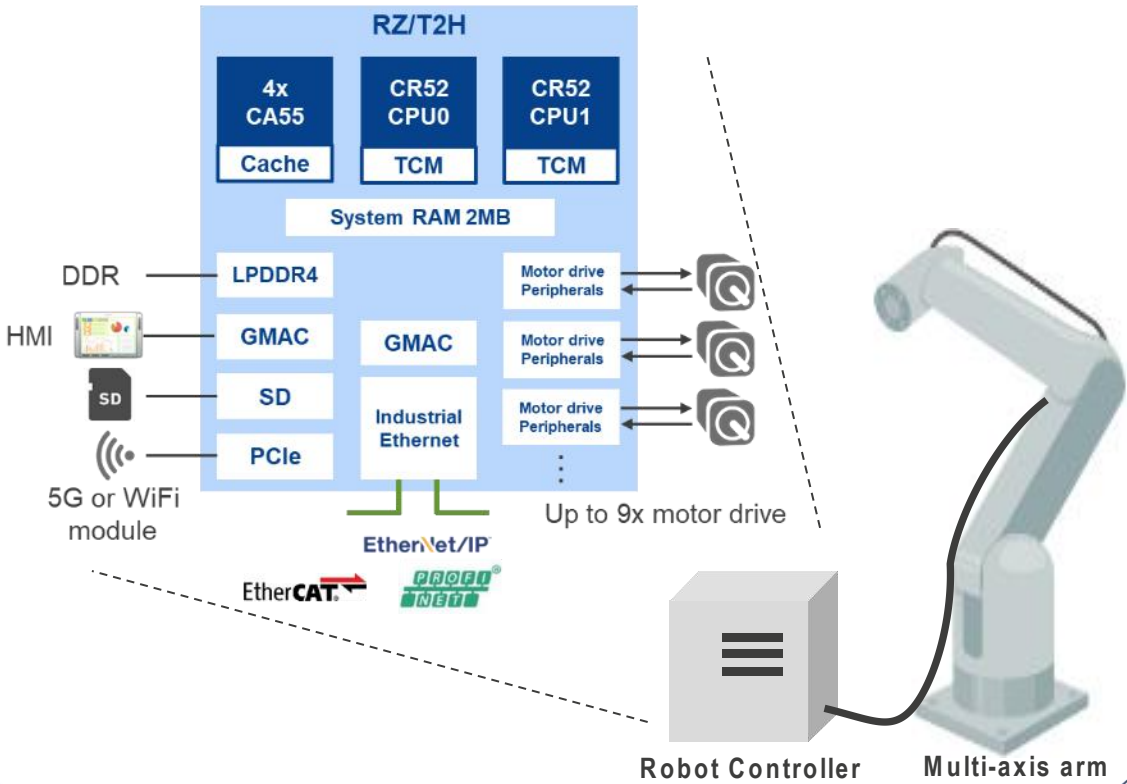
INDUSTRIAL ROBOTIC

INTEGRATED SERVO + MOTION CONTROL SOLUTION



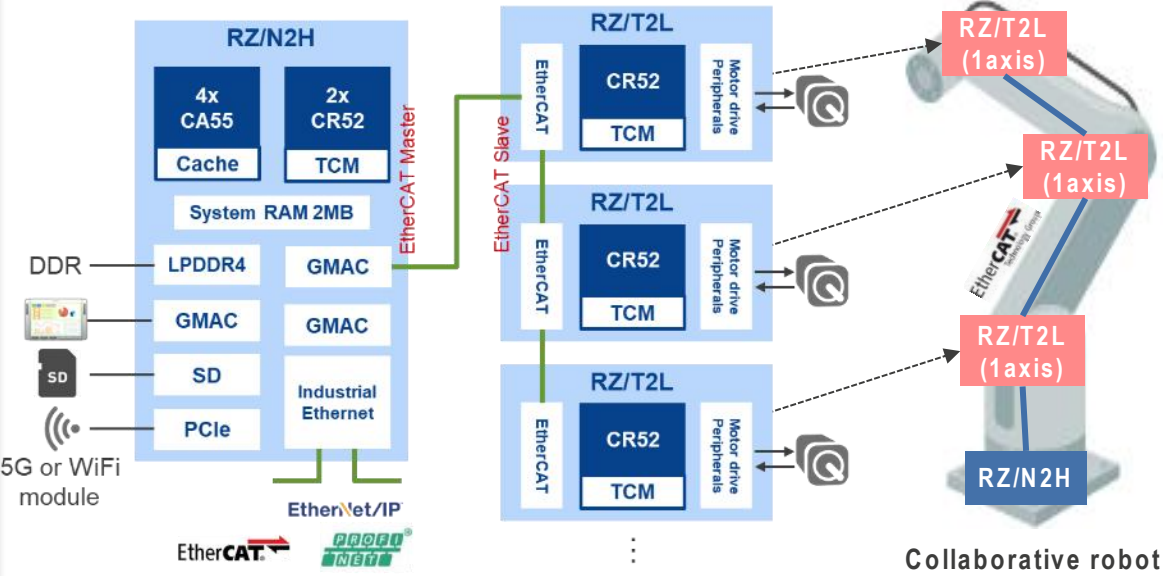
RZ/T2H one chip implementation (ex. Industrial Robot)

RZ/T2H realize motion control and motor control in one chip



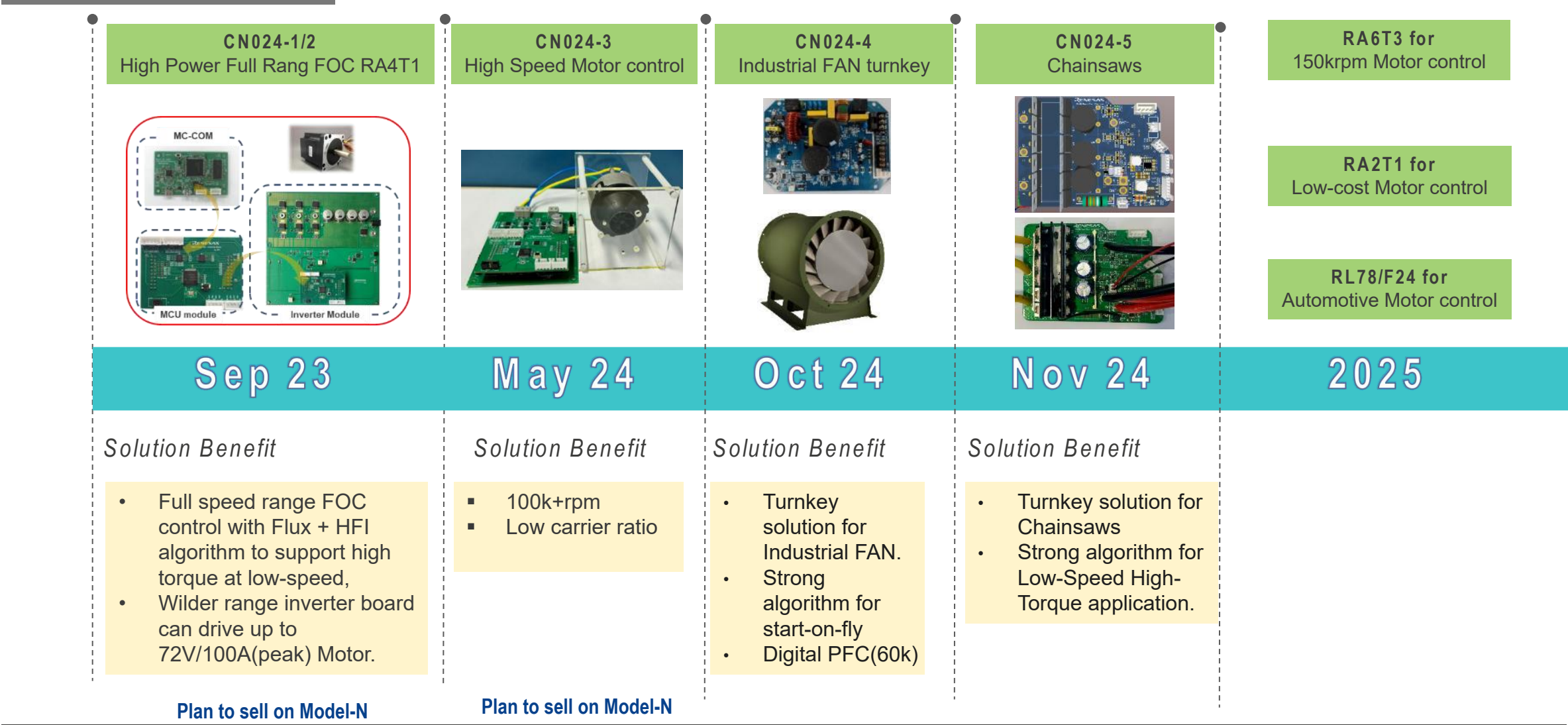
RZ/N2H and RZ/T2L two chip implementation (ex. Collaborative robot)

RZ/N2H realize motion control and Network master
RZ/T2L realize each axis control



MOTOR SOLUTION ROADMAP

COVERED WILDER MOTOR USE CASES IN ROBOTIC



RENESAS MOTOR CONTROL ALGORITHM RESOURCES

Magnetic Encoder

[Motor and Inverter Control Solutions | Renesas](#)

[Resolver Solution](#)

[Induction Motor Solution](#)

[Magnetic Encoder Solution](#)

[Brushless DC Motors \(Permanent Magnet Synchronous Motors\) Solution](#)

[Digital Control Solution for 3-Level Inverters](#)

[Whole Speed Range Sensorless Motor Solution](#)

[Motor Control Algorithms | Renesas](#)

Control Algorithm	Motor Type	Torque Controllability	Feedback Sensors	MCU Processing Requirements
TRIAC Control	Universal / DC	No Torque Control	Not Required	Not Required
PWM Chopper Control	Universal / DC	No Torque Control	Not Required	8 bit
V/F Control	AC	Poor Dynamic Torque Control	Not Required	8-16 bit
Trapezoidal Drive Control	BLDC	High-Harmonic Loss, High-Torque Ripple	Hall Sensor	8-16 bit
Sinusoidal Drive Control	BLDC, AC	Low-Harmonic Loss, Low-Torque Ripple	Hall Sensor, Encoder	16-32 bit
Simplified Vector Control	BLDC, AC	Low-Harmonic Loss, Low-Torque Ripple	Hall Sensor, Encoder	16-32 bit
Vector Control	BLDC, AC	Low-Harmonic Loss, Low-Torque Ripple	Hall Sensor, Encoder	32 bit
Sensorless Trapezoidal Drive Control	BLDC	High-Harmonic Loss, High-Torque Ripple	Not Required (Back-EMF)	16 bit
Sensorless Vector Control	BLDC, AC	Low-Harmonic Loss, Low-Torque Ripple	Not Required	32 bit

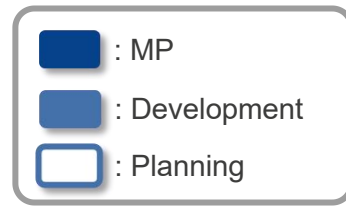
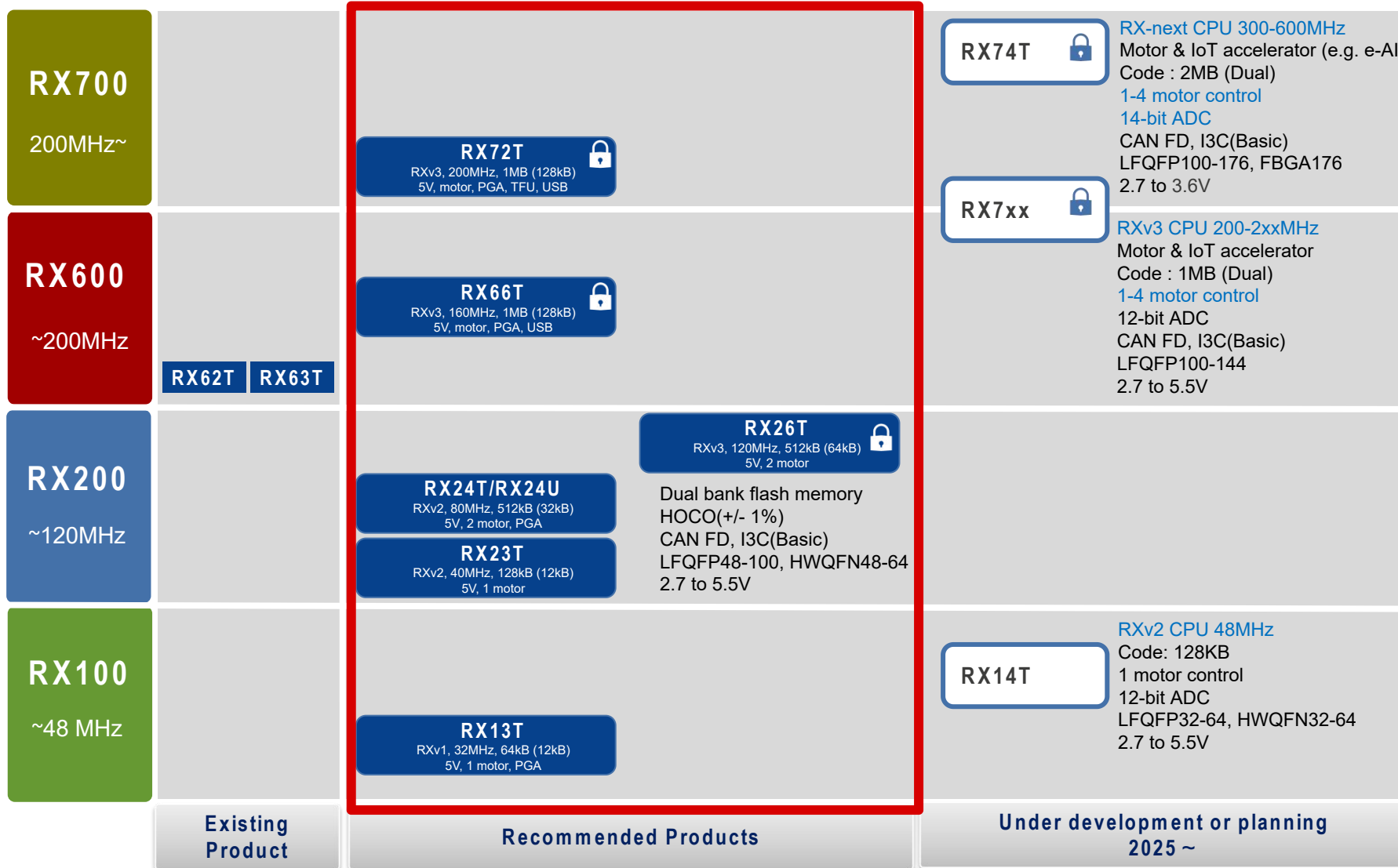
RA FAMILY PRODUCT PORTFOLIO AND MOTOR DEVICES

RA8	RA8P1 1GHz Cortex-M85 + 250MHz M33, ~1MB MRAM, Ethos-U55, Giga Ethernet/TSN, OSPI, RSIP-E51A				Status: Apr 2025
	RA8D1 480MHz Cortex-M85, ~2MB Flash, USBHS/FS, Ethernet, OSPI, CAN-FD, Camera I/f, RSIP-E51A	RA8D2 1GHz Cortex-M85 + 250MHz M33, ~1MB MRAM, USBHS/FS, Giga Ethernet/TSN, HMI/Gfx, RSIP-E51A	RA8E2 480MHz Cortex-M85, 1MB Flash, USBFS, OSPI, CAN-FD, Graphic LCD, 2DG		
	RA8M1 480MHz Cortex-M85, ~2MB Flash, USBHS/FS, Ethernet, OSPI, CAN-FD, Camera I/f, RSIP-E51A	RA8M2 1GHz Cortex-M85 + 250MHz M33, ~1MB MRAM, USBHS/FS, Giga Ethernet/TSN, OSPI, RSIP-E51A	RA8E1 360MHz Cortex-M85, ~1MB Flash, USBFS, Ethernet, OSPI, CAN-FD, Camera I/f		
RA6	RA6M3 120MHz Cortex-M4, ~2MB Flash Ethernet, USBHS, CAN, Graphics, JPEG, TFT LCD, SCE7	RA6M6			
	RA6M2 120MHz Cortex-M4, ~1MB Flash Ethernet, USBFS, CAN, SCE7	RA6M5 200MHz Cortex-M33, ~2MB Flash TrustZone, Ethernet, USBFS+ USBHS, CAN-FD, Octa SPI, SCE9	RA6L1	RA6E1 200MHz Cortex-M33, ~1MB Flash TrustZone, Ethernet, USBFS, CAN	
	RA6M1 120MHz Cortex-M4, 512KB Flash USBFS, CAN, SCE7	RA6M4 200MHz Cortex-M33, ~1MB Flash TrustZone, Ethernet, USBFS, CAN, Octa SPI, SCE9		RA6E2 200MHz Cortex-M33, ~256KB Flash TrustZone, USBFS, CAN FD, I3C	
RA4	RA4M1 48MHz Cortex-M4, 256KB Flash USBFS, CAN, Seg. LCD, CTSU Touch Sensing, 14bit SAR ADC, SCE5	RA4M3 100MHz Cortex-M33, ~1MB Flash TrustZone, USBFS, CAN, SCE9	RA4L1 80MHz Cortex-M33, ~512KB Flash Low-power, LCD, Touch, USB, CAN-FD, RSIP3	RA4E1 100MHz Cortex-M33, ~512KB Flash TrustZone, USBFS, CAN	
		RA4M2 100MHz Cortex-M33, ~512KB Flash TrustZone, USBFS, CAN, SCE9	RA4L2 80MHz Cortex-M33, ~256KB Flash Low-power, USB, CAN-FD, RSIP3	RA4E2 100MHz Cortex-M33, 128KB Flash TrustZone, USBFS, CAN FD, I3C	
RA2	RA Family Product Portfolio as of Apr. 2025		RA2L1 48MHz Cortex-M23, ~256KB Flash 48-100pin, CAN, Touch, 5V	RA2E1 48MHz Cortex-M23, ~128KB Flash 25-64pin, Touch, 5V	
			RA2L2 48MHz Cortex-M23, 128KB Flash 32-64pin, I3C, 125C	RA2E2 48MHz Cortex-M23, ~64KB Flash 16-24pin, I3C, 5V, 125C	
				RA2E3 48MHz Cortex-M23, ~64KB Flash 32-48pin, 5V	
RA0	RA Family Product Portfolio as of Apr. 2025		RA0Lx	RA0E2 32MHz Cortex M23, 64~128KB Flash 32-64pin, 125C, 5V	
			RA0L1 32MHz Cortex M23, 32~64KB Flash 20-48pin, 125C, 5V, Touch	RA0E1 32MHz Cortex M23, 32~64KB Flash 16-32pin, 5V	
				RA0Ex	
				RA0Tx	RA0Ax
Mainstream Line			Low Power	Entry Line	Motor Control
					Rich Analog / Metering / Wireless

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RX PRODUCT ROADMAP AND MOTOR DEVICES

RX-T (FOR MOTOR / INVERTER CONTROL)



The RX-T series has been adopted mainly in air conditioners and industrial inverters and is highly evaluated in the field of inverter control. In recent years, it is getting adopted in power supply control applications such as UPS, solar inverters, and EV chargers as well.

Please look forward to the future development of the Renesas RX family expansion!

RZ FAMILY PRODUCT PORTFOLIO AND MOTOR DEVICES

Linux / Android™ / Multi-OS with RTOS

RTOS

Vision AI

RZ/V Series

RZ/V2M

1.0GHz Dual-core Cortex-A53,
DRP-AI(576-MAC), 4K-ISP

RZ/V2N

1.8GHz Quad-core Cortex-A55,
DRP-AI3(2K-MAC), 4K-ISP, 3D-GPU

RZ/V2H

1.8GHz Quad-core Cortex-A55,
DRP-AI3(4K-MAC), 4K-ISP, 3D-GPU

RZ/V2MA

1.0GHz Dual-core Cortex-A55, DRP-AI(576-MAC), OpenCV Accelerator

RZ/V2L

1.2GHz Dual-core Cortex-A55,
DRP-AI(576-MAC), 3D-GPU

Industrial Network

RZ/N Series

RZ/N1D

500MHz Dual-core Cortex-A7, 125MHz Cortex-M3, Industrial Ethernet

RZ/N2H

1.2GHz Quad-core Cortex-A55, 1GHz Dual-core Cortex-R52, 6-axis control

RZ/N1L

125MHz Cortex-M3, Industrial Ethernet

RZ/N1S

500MHz Cortex-A7, 125MHz Cortex-M3, Industrial Ethernet

RZ/N2L

400MHz Cortex-R52, Industrial Ethernet

Real-time Control

RZ/T Series

RZ/T2H

1.2GHz Quad-core Cortex-A55, 1GHz Dual-core Cortex-R52, 9-axis control

RZ/T2L

800MHz Cortex-R52, 2-axis motor control, EtherCAT

RZ/T2ME

800MHz Dual-core Cortex-R52, 2-axis motor control, Industrial Ethernet, OTFD

RZ/T1

600MHz Cortex-R4, 150MHz Cortex-M3, 1-axis motor control, Industrial Ethernet

RZ/T2M

800MHz Dual-core Cortex-R52, 2-axis motor control, Industrial Ethernet

IoT Edge

RZ/G Series

RZ/G3S

1.1GHz Cortex-A55, 250MHz Dual-core Cortex-M33, LPDDR4/DDR4

RZ/Five

RISC-V, 1.0GHz AX45MP, DDR4/3L, GbEthernet, CAN-FD

RZ/G1H

1.4GHz Quad-core Cortex-A15 + Cortex-A7, DDR3, 3DG, H.264

RZ/G2H

1.5GHz Quad-core Cortex-A57 + Cortex-A53, LPDDR4, 3DG, H.264/5

RZ/G2L

1.2GHz Dual-core Cortex-A55, DDR4/3L, 3DG, H.264, CAN-FD

RZ/G1M

1.5GHz Dual-core Cortex-A15, DDR3L, 3DG, H.264

RZ/G2M

1.5GHz Dual-core Cortex-A57 + Cortex-A53, LPDDR4, 3DG, H.264/5

RZ/G2LC

1.2GHz Dual-core Cortex-A55, DDR4/3L, 3DG, CAN-FD

RZ/G1N

1.5GHz Dual-core Cortex-A15, DDR3L, 3DG, H.264

RZ/G2N

1.5GHz Dual-core Cortex-A57, LPDDR4, 3DG, H.264/5

RZ/G2UL

1.0GHz Cortex-A55, DDR4/3L, CAN-FD, ADC

RZ/G1E

1.0GHz Dual-core Cortex-A7, DDR3, 3DG, H.264

RZ/G2E

1.2GHz Dual-core Cortex-A53, DDR3L, 3DG, H.264/5

HMI

RZ/G Series

RZ/A Series

RZ/A1LU

400MHz Cortex-A9, 3MB RAM, LCDC, JPEG, Ethernet, USB

RZ/A1H

400MHz Cortex-A9, 10MB RAM, LCDC, JPEG, Ethernet, USB

RZ/A3UL

1.0GHz Cortex-A55, DDR4/3L, LCDC, GbEthernet, USB

RZ/A1L

400MHz Cortex-A9, 3MB RAM, LCDC, Ethernet, USB

RZ/A1M

400MHz Cortex-A9, 5MB RAM, LCDC, JPEG, Ethernet, USB

RZ/A2M

528MHz Cortex-A9, 4MB RAM, LCDC, JPEG, MIPI-CSI, Ethernet, USB

RZ/A1LC

400MHz Cortex-A9, 2MB RAM, LCDC, Ethernet, USB

FUSA SOLUTION OVERVIEW



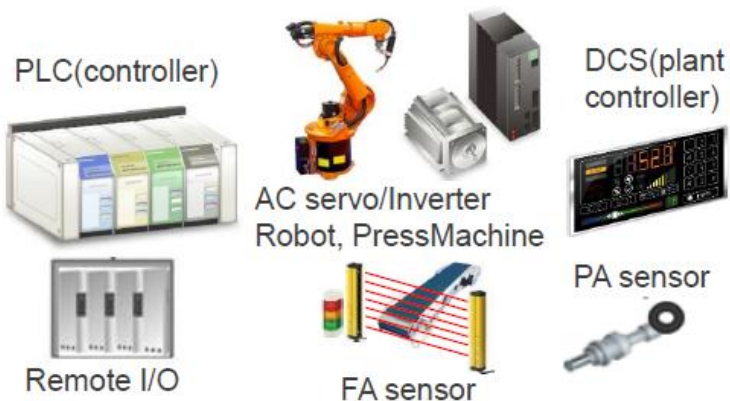
MAJOR FUNCTIONAL SAFETY STANDARD

Segment Example	Safety standard (Safety level)	Renesas Deliverable		
		Renesas MCU/MPU	Certify body	Safety level definition
Automotive	ISO 26262 (ASIL A-D)	RH850	NA	ASIL
Industry	IEC 61508 (SIL 1 to 4)	RX111/113/130/140 RXv2, and v3 core RA CM4, CM23, CM33 RZ/T2x, RZ/N2x	TUV	SIL
Home Appliance	IEC 60730 (Class A/B/C)	RL78 RA RX Synergy	VDE (Class B/C by TUV coming soon)	Class

IEC61508 STANDARD GOING BROADER

FROM INDUSTRIAL AUTOMATION TO OTHER SEGMENTS

FA and PA (IEC61508, others)



Motor assist cycle (EN15194)



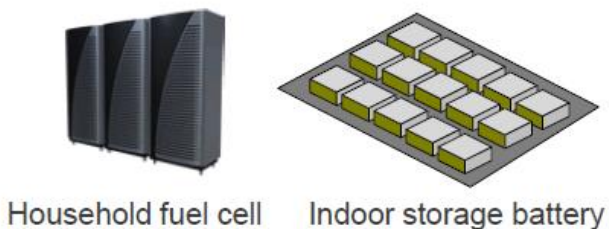
BA



Automatic Door (EN16005)



Battery System (IEC62133, others)



Service Robot (ISO13482)



Printer



FUNCTIONAL SAFETY SOLUTION OVERVIEW

EXAMPLE HOW OUR BUILDING BLOCKS REALIZE SAFETY SYSTEM

- Customer can select the required solution blocks depending on their knowledge.

Use Case: Safety drive + safety network system
SIL3, HFT=1

1. Self-Test Software Kit

Diagnostic Software for permanent failure of CPU, ROM, RAM inside the MCU



2. SIL3 System Software Kit

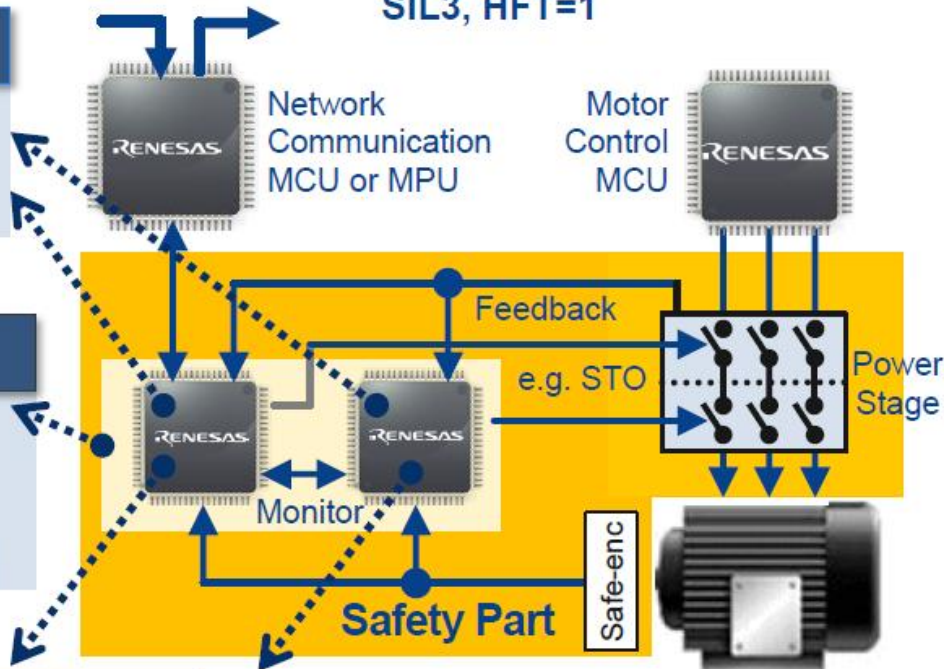
Functional Safety Platform Software for dual MCU system equipped with MCU diagnosis, scheduler, partitioning function



3. FSoE Application Software Kit

4. PROFIsafe Application Software Kit

Safety protocols for slave devices



5. Reference Document

Guideline document to obtain IEC61508 certification

- Sample documents to be submitted to certification body & preparation guide
- Technical documents necessary for safety part development such as input and output circuit diagnosis, power-supply monitoring

6. Reference Hardwares

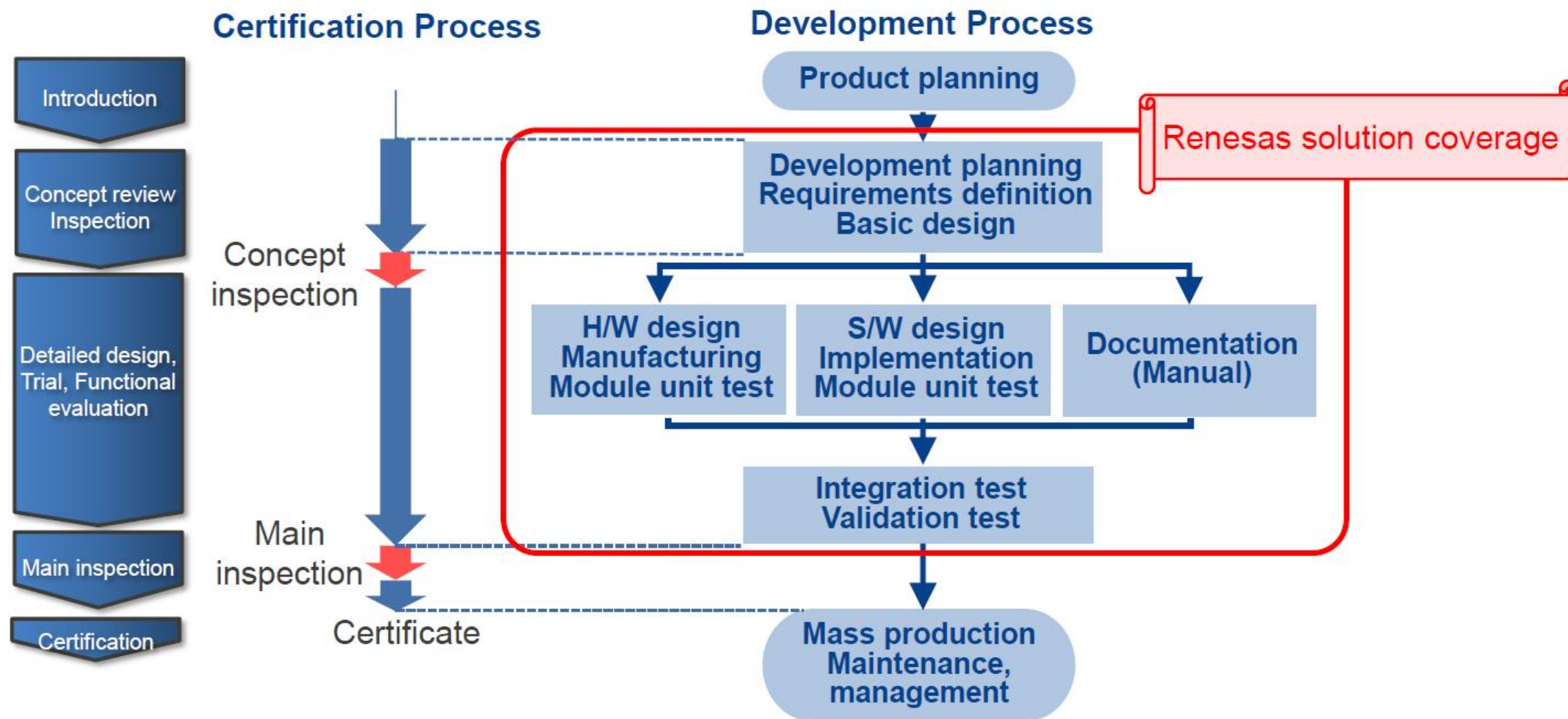
Dual MCU configuration Evaluation Board

7. IEC61508 Certification Kit for RX Compilers

Document set certifying functional safety compliance of Renesas compiler "CC-RX"



FUNCTIONAL SAFETY SYSTEM DEVELOPMENT TO CERTIFICATION



MOTOR CONTROL SYSTEM WITH INDUSTRIAL AND FUNCTIONAL SAFETY

System Overview

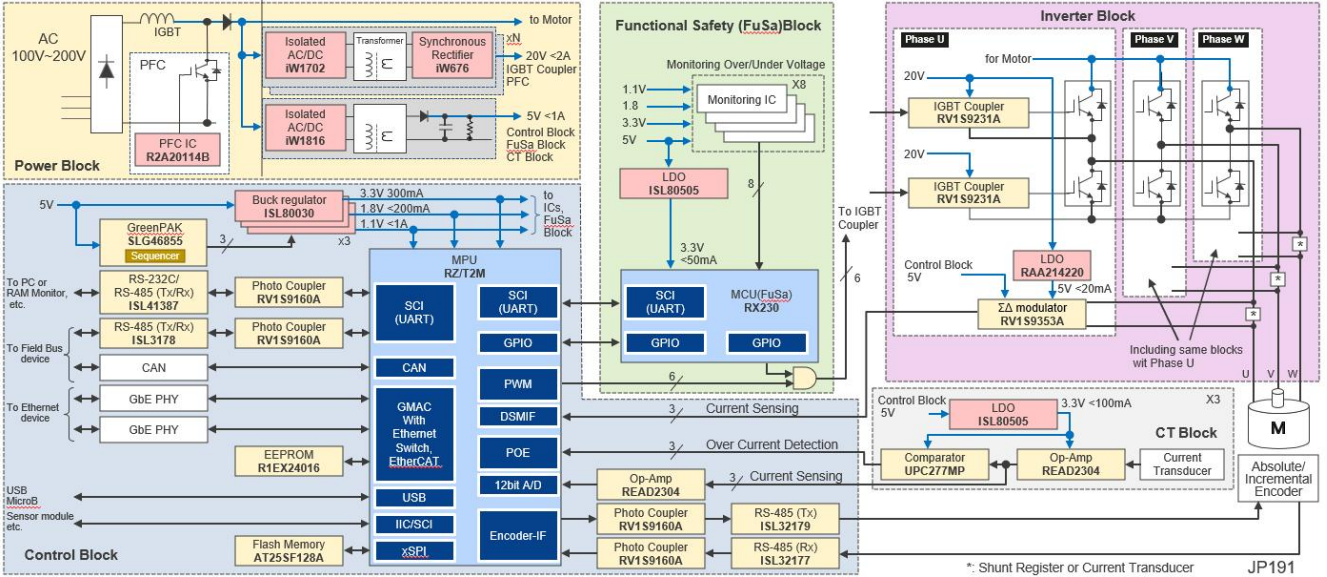
Renesas provides a comprehensive system for industrial motor control, integrating an MPU and MCU for mutual monitoring, along with power supply ICs, delta-sigma ($\Delta\Sigma$) modulators, and other essential devices. This combination of components enables a streamlined and high-performance approach to motor control, industrial networking, and functional safety, ensuring efficient and reliable operations in industrial environments.

System Benefits

- Configures a high-precision system by combining the MPU with peripheral devices, reducing the need for external components.
- Utilizes a dual CPU core in the MPU to achieve high-precision motor control and industrial Ethernet communication.
- Supports major industrial Ethernet protocols such as EtherCAT, Ethernet/IP, PROFINET IRT, and OPC Unified Architecture (OPC UA) with a simple configuration requiring minimal external components.
- Simplifies the creation of a functional safety system with an MPU and one MCU for mutual monitoring.
- Includes the RZ/T2M Motor Solution Kit for comprehensive evaluation and development.
- Features the prepared SBEV-RZ/T2M evaluation board from Shimafuji Electric Inc., allowing for the development and evaluation of software for best-in-class real-time control applications.

Target Applications

- AC servo motors
- Inverters
- Industrial robots



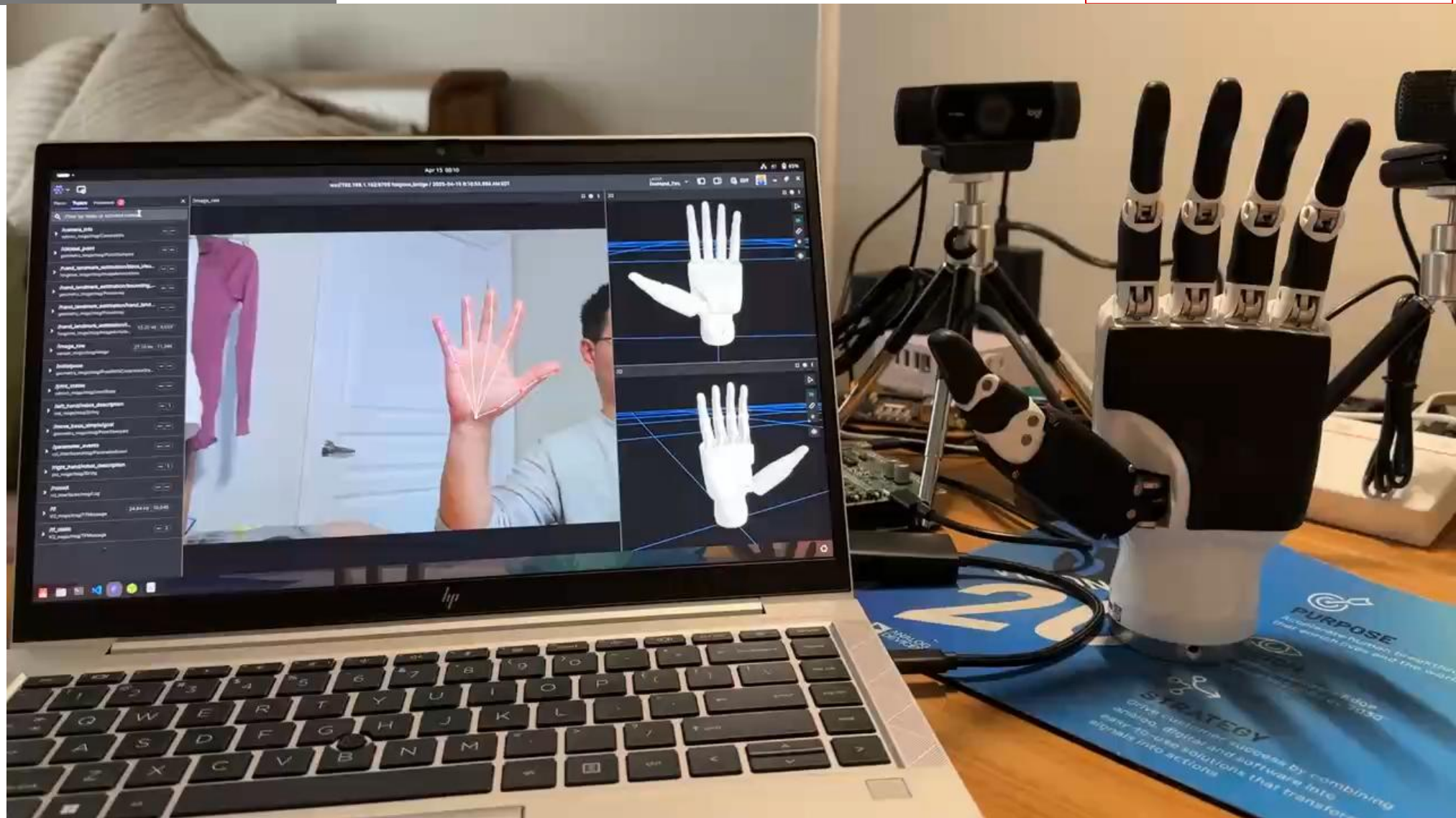
Category	P/N	Key Features EP
EP	RZ/T2M R9A07G075M28GBG#AC0	Arm®-based High-end 32-bit MPU, Real-time control + Industrial Ethernet, FuSa
	RX230 R5F52318ADLA#20	32-bit MCU with Enhanced DSP, FPU and CTSU
Analog	SLG46855V	GreenPAK™ Programmable Mixed-signal Array. (Reference Design: SLG7RN46468)
	RV1S9231ACCSP-10YC#KC0	IGBT Gate Drive Photocouplers
	RV1S9353ACCSP-120V#KC0	Optically Isolated $\Delta\Sigma$ Modulator
	ISL3178EIBZ	High ESD Protected RS-485/RS-422 Transceivers
Power	iW1816-20-S07	PWM Converter with 800V Bipolar Power Transistor
	iW1702-00B	45W Digital AC/DC PWM Controller Optimized for 9V+ Applications
	ISL80030FRZ-T	2.7V to 5Vin, 3A Sync Buck
	RAA2142204GP3#JA0	20V Low dropout regulators 150mA

AI AND HUMANOID ROBOTIC PLATFORM



AI DEX-HAND SOLUTION DEMO

[Click to play the video](#)



RENESAS AI PLATFORM

- Renesas **creates leading semiconductor solutions** that spark **innovation** for a connected world, building a trusted brand we can be proud of.

Trusted



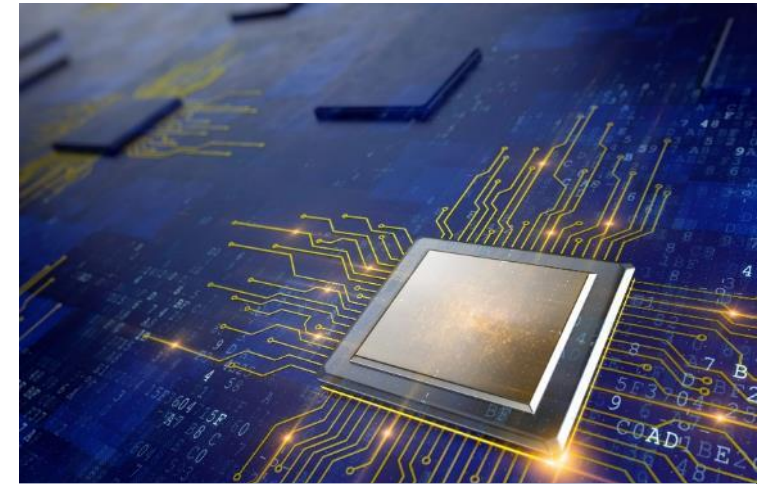
Scalable & Auto. proven platform
➔ High quality and full ASIL D SoC

Innovative



Most efficient Deep Learning IP
➔ Extending toward generic compute

Open



One SW Platform & Tool chain
➔ Seamless reuse & devt efficiency

Renesas Autonomy enables embedded intelligence for the future of driving

VISION AI SCALABILITY

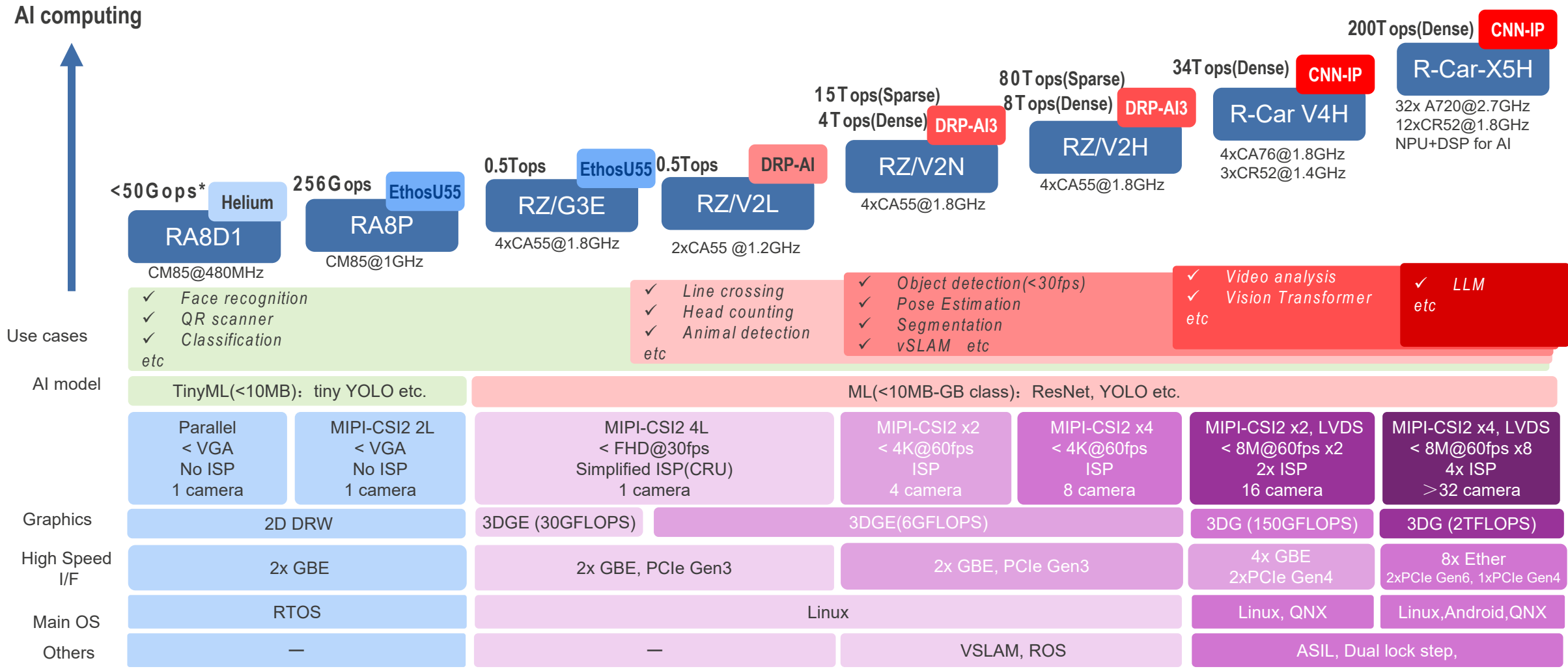
■ Vision AI scalability from RA to RZ/V to RCAR covering variety of use cases and conditions

Low BOM cost with RTOS environment

Easy AI development with Linux ecosystem

High performance and multi vision AI

Ultra High performance for AI with surround view



*unofficial value from rough estimation with AI performance

JOINTS CONTROL AND JOINT MOTOR ARCHITECT

High-Precision Dual Encoders •

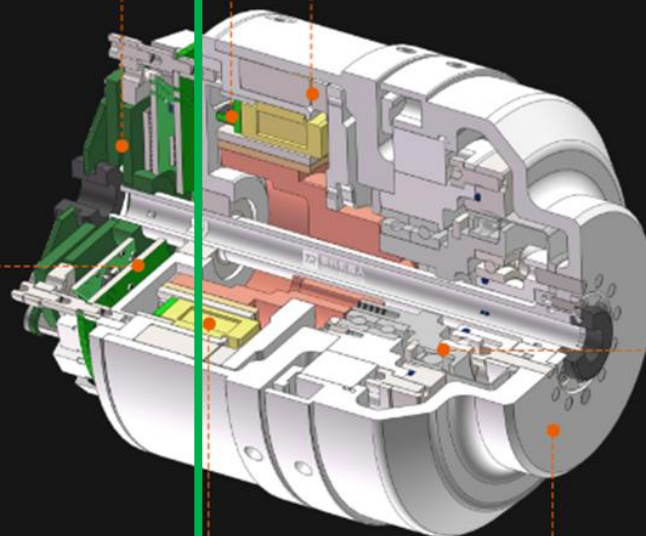
- 17-bit high resolution, continuously pursuing precision.
- Ultra-thin design, compact structure.
- High resolution, high precision.
- Incremental & Single-turn absolute value encoder.

Servo Drive •

- Supports EtherCAT/CANopen optionally.
- High-performance all-digital servo drive.
- Ultra-thin intelligent integration, smooth operation, and quick response.
- Outstanding performance, precise control.

Frameless Torque Motor •

- Low torque ripple, low inertia.
- High torque density output.
- High precision, high rigidity, and faster response speed.



Permanent Magnet Brake*

- No sway during shutdown, no jitter during startup, and capable of full-load zero-speed start.
- Outstanding safety braking, instant hold for safety collision.
- Compact structure, low noise, high safety, long working lifespan.
- Minimal torque variation, allowing instant braking of the load even in abnormal conditions.

Precision Harmonic Reducer

- Compact size, lightweight.
- Smooth torque output.
- Low noise, low vibration.
- Stable and efficient performance.

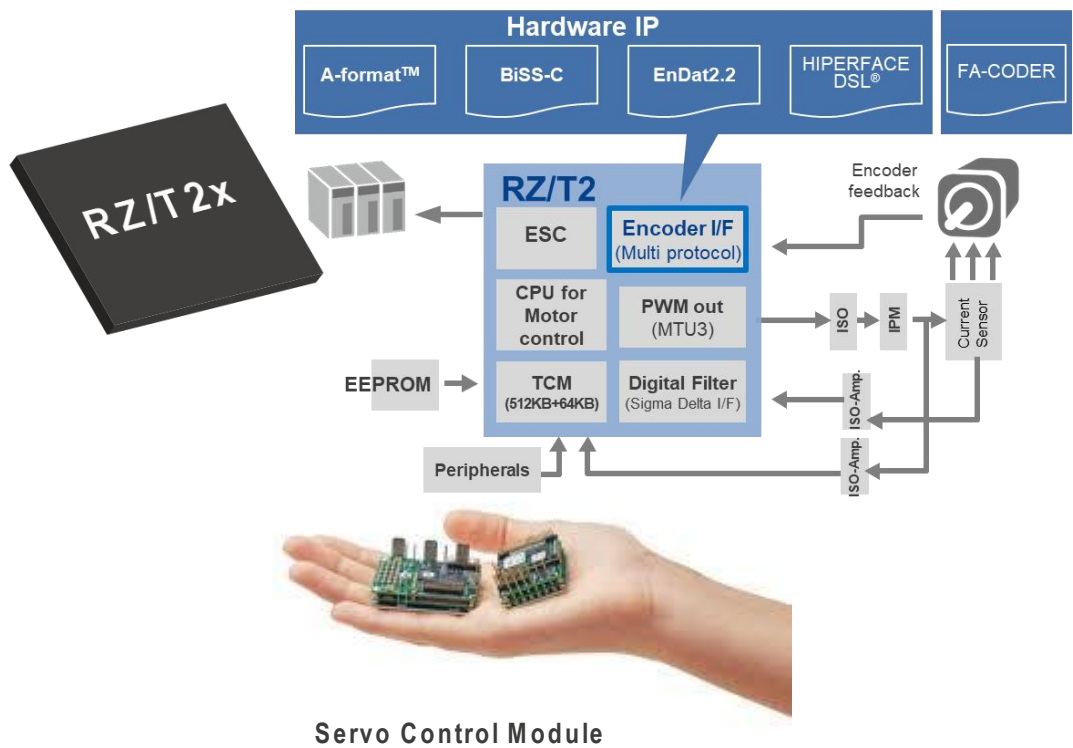
Force Sensor (Optional)

- According to the requirements of the use scenario, higher precision motion control and more complex flexibility operations can be achieved by optionally installing force sensors.

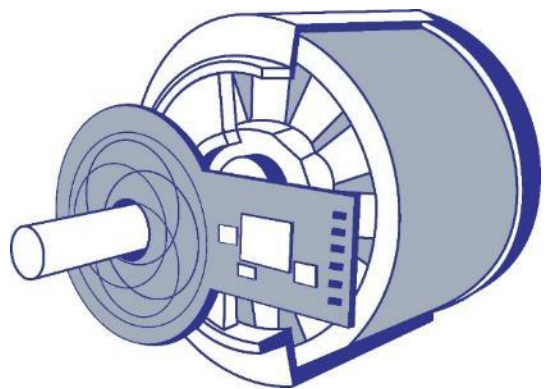
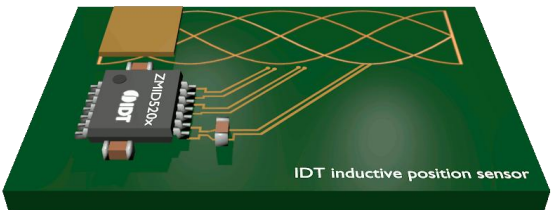
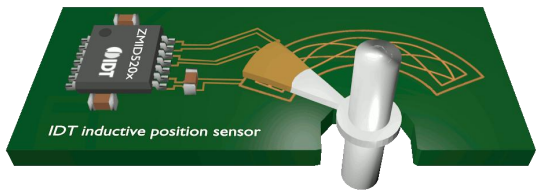
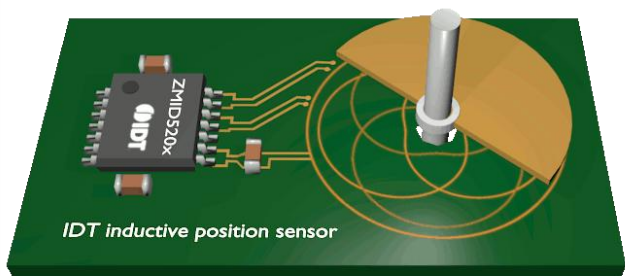
*Certain models have optional brakes; refer to each model's specifications for details.

JOINTS CONTROL AND JOINT MOTOR SOLUTIONS

RZ/T ALL IN ON SOLUTION WITH INVERTER FOR JOINT CONTROL

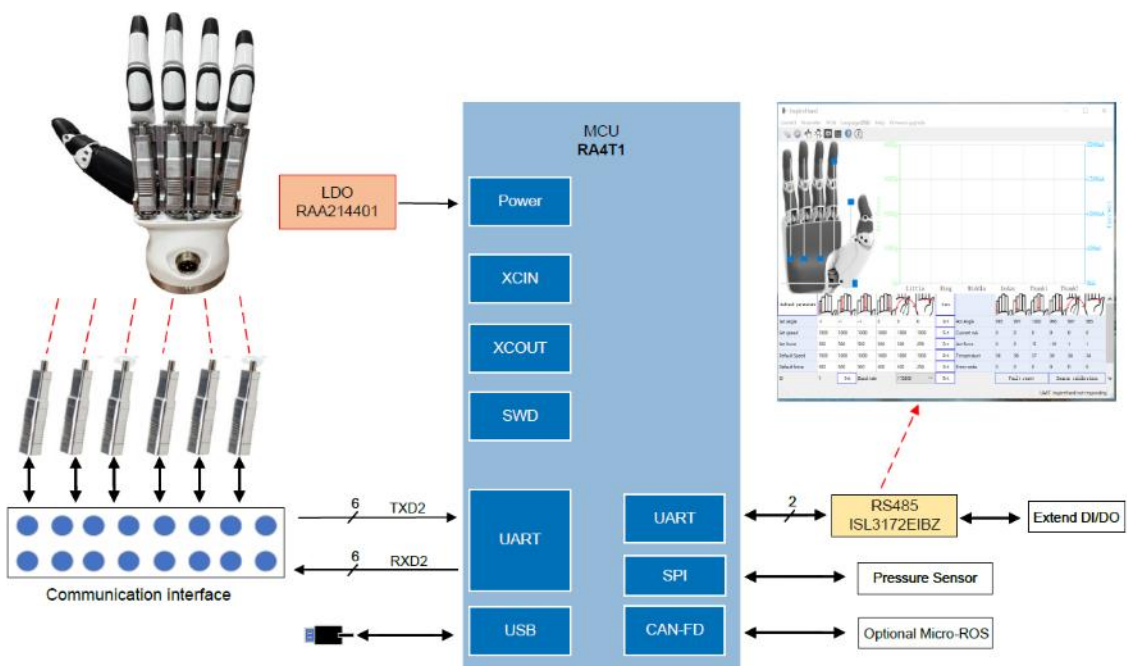


Best Position Sensor for Joint Encoder Motor Total stray field immunity ISO 11452-8 compliant

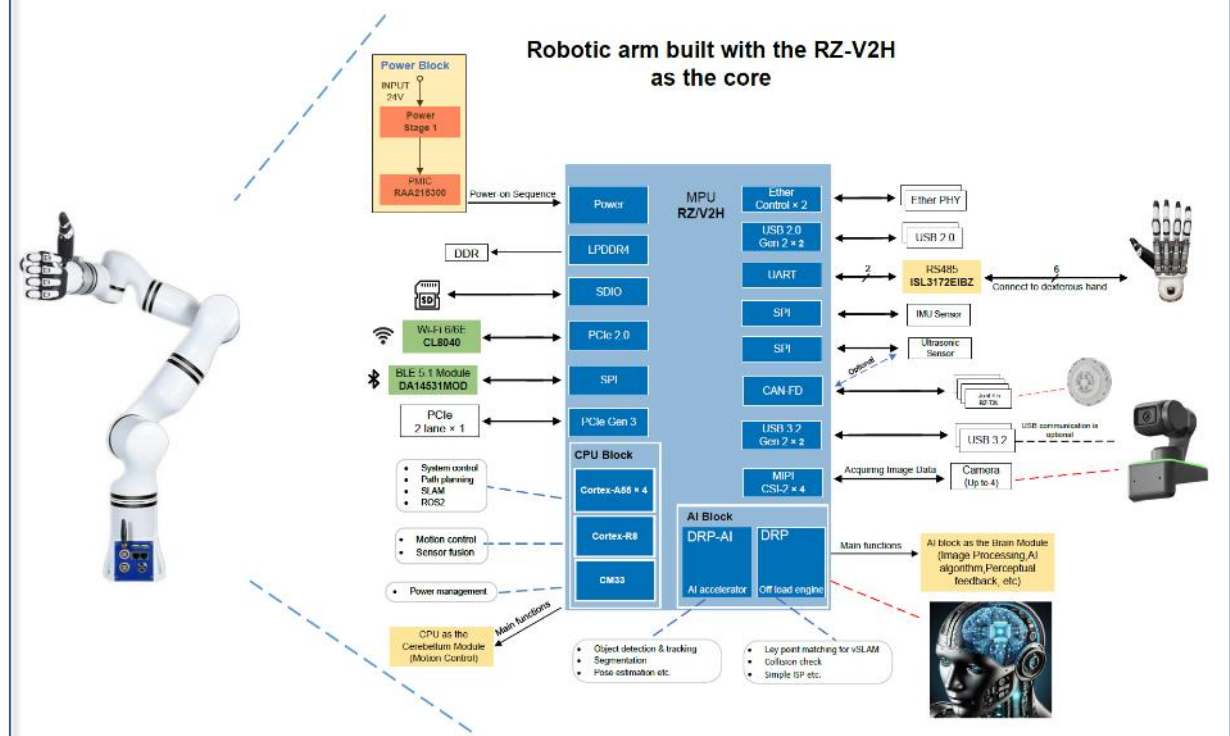


DEX-HAND AND ARM SOLUTIONS

The dexterous hand Solution



The Hand+ARM+Vision AI solution



DEXTEROUS HAND SOLUTION—RELEASE LIST



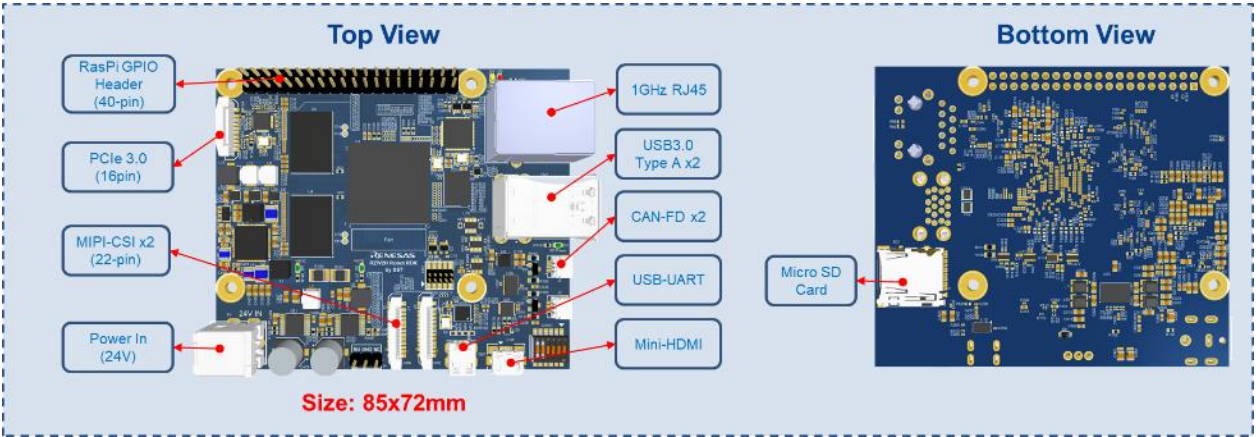
- Core Value:** Renesas Dex-Hand platform
- ❖ Dex-Hand design Kit: MCU for Control, SoC for AI
 - ❖ Dex-Hand MCU reference code with Micro-ROS and CAN-FD
 - ❖ Vision AI hand initiative with Renesas low-power 8T MPU(RZ/V2H)

Num	Release Result		Link
1	Firmware	RA4T1 for Exterous Hand Project	
2		RA4T1 for MicroROS Sample Project	
3	Hardware	RA4T1 for Exterous Hand Schematic	
4		RA4T1 for Exterous Hand PCB	
5	Manual	Quick Startup Guide	
6	Video	Video for comparing with product and demo	

ROBOT DEVELOPMENT KIT

System Overview

- This solution equipped with Renesas new generation MPU RZ/V2H for AI application, which has AI inference processing performance of up to 8TOPS with multi-core CPU to run multiple OS simultaneously for high performance AI image processing.
- It also equipped with many interfaces that making it is suitable for development and integration into a variety of applications especially for robotics.
- Support the latest version Yocto 5.0 (Scarthgap) and Ubuntu 22.04 / 24.04. All the design files can be delivered to customers to accelerate their development.

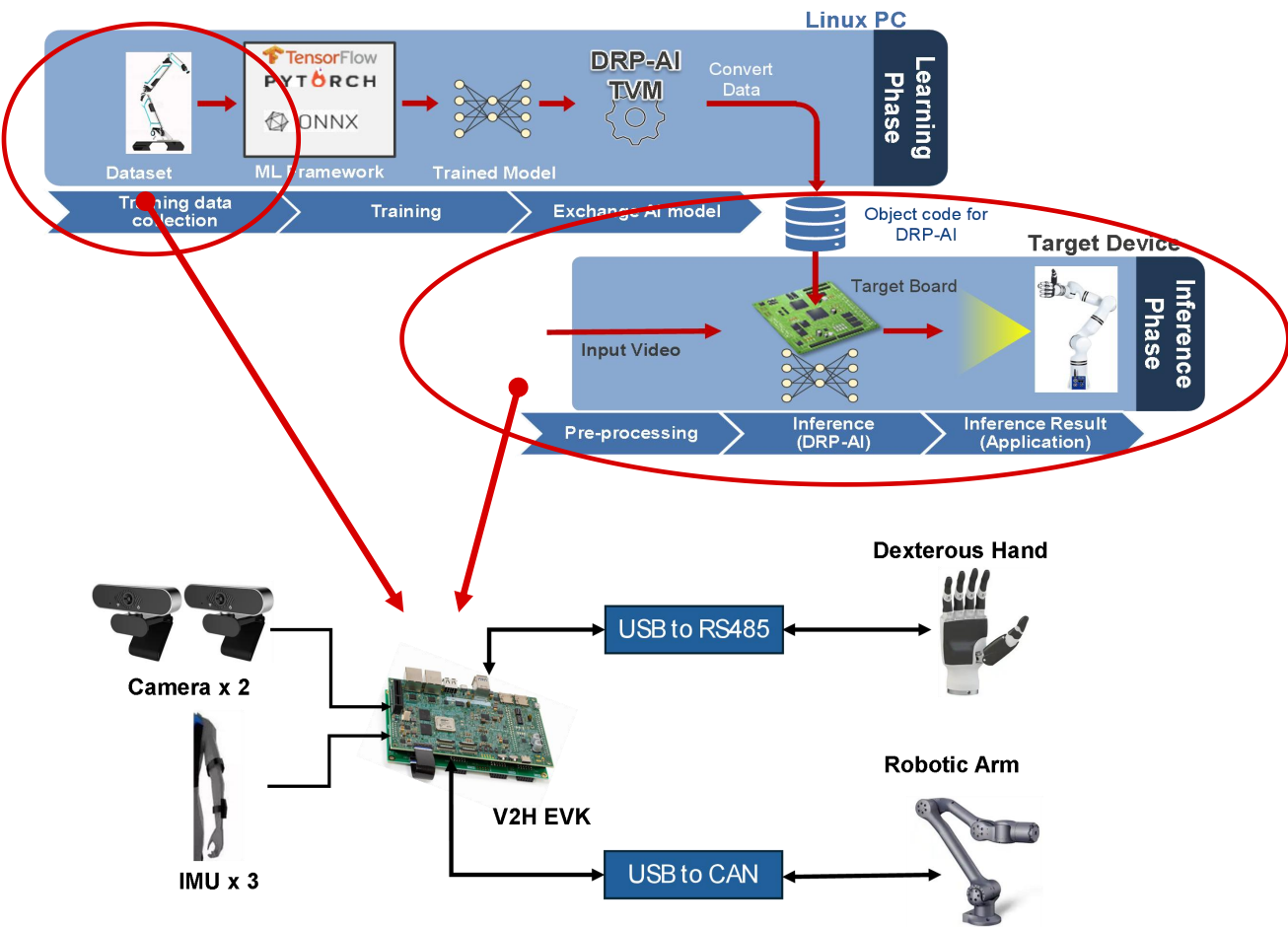
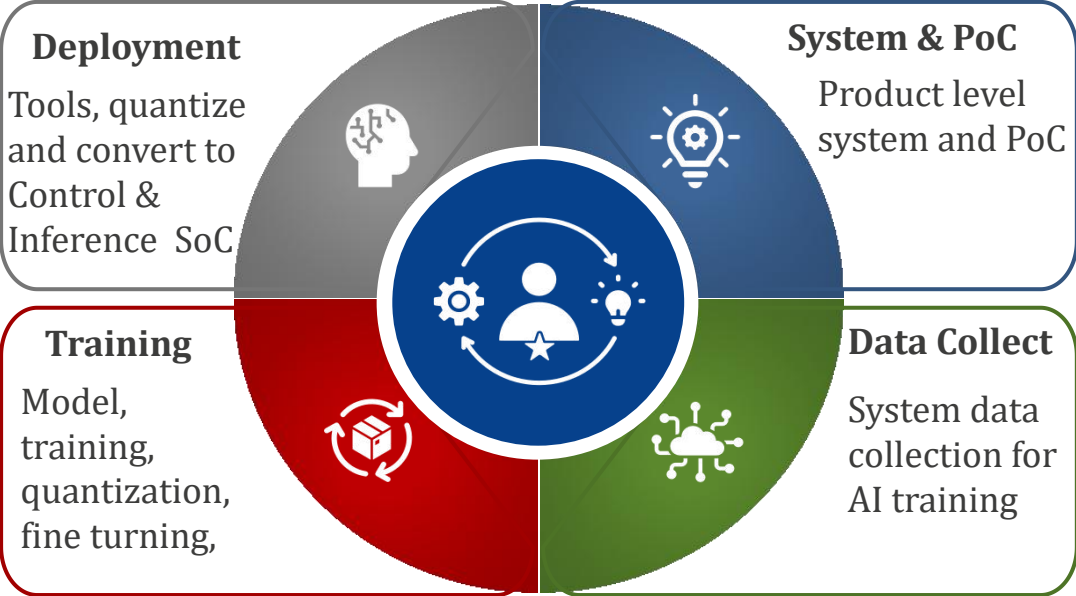


Specifications	
MPU	Renesas RZ/V2H, AI-MPU
	Cortex-A55 Quad 1.7GHz Cortex-R8 Dual 800MHz
Memory	LPDDR4 1600MHz (8GB) x 2
Ethernet	10/100/1000 Base-T RJ45
Video In	MIPI-CSI 4-Lane x2 (22-pin/0.5mm)
Video Out	Mini HDMI
PCIe	PCIe 3.0 End Point (16-pin/0.5mm)
USB	USB3.0 Type A x2
CAN	CAN-FD x2
SD Card	Micro SD
GPIO	40-pin RasPi GPIO Header
Power In	DC Jack (24V/2A)
Size	85x72mm

ROBOTIC AI FRAMEWORK

Core Value: Renesas Robotic AI platform

- ❖ Renesas Robotic AI Kit: RZ/V2H to RCAR
- ❖ Robotic demo system with HW/SW solutions
- ❖ Robotic AI training and deployment system



The image features the Renesas logo in a bold, blue, sans-serif font, centered horizontally. The background is a grayscale, isometric illustration. In the foreground, a detailed circuit board is shown with various components like chips and connectors. In the background, a city skyline with several tall skyscrapers is visible, creating a sense of technology integrated with urban infrastructure.

RENESAS

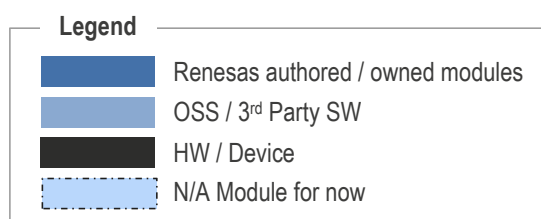
APPENDIX

SW ARCHITECT AND SOLUTION ONE PAGE

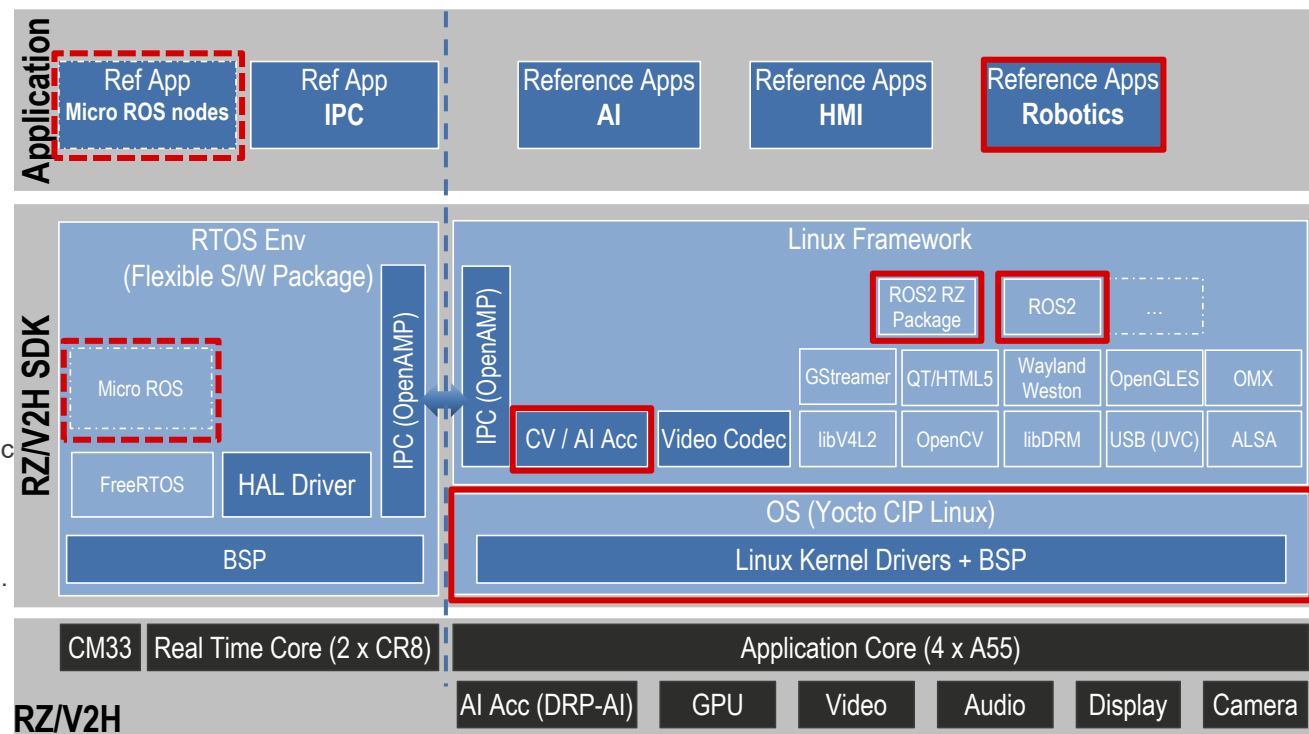


RZ/V2H SW STACK OVERVIEW AND ANALYSIS

SIMPLIFIED FOR ROBOTICS ECOSYSTEM INVESTIGATION



- ❖ **Update Yocto (toolset to build Linux distro)**
 - To integrate the latest Robotics middleware, ROS2 Humble or later (via the meta-ros2 layer), along with a new kernel, Yocto 5.0 (Scarthgap) is required.
 - The new release may also include a new kernel (6.x), which is in high demand among non-automotive customers. (Currently, Kernel 5.10 is in use.)
 - RZ/V2H are still using Yocto 3.1 (Dunfell). Official update schedule is currently unclear.
- ❖ **Support Ubuntu 22.04 / 24.04**
 - Designated OS for latest ROS2 release for optimal compatibility and performance.
- ❖ **Update ROS2 (M/W framework for robotic systems development)**
 - RZ/V2H already provides the ROS2 support package, but it is based on outdated ROS2 Galactic and containerized ROS2 Humble. New LTS version of ROS2 (Humble or Jazzy) shall be supported natively after updating the Yocto toolset.
- ❖ **Provide more ROS2 support packages and nodes (ROS2 RZ Package)**
 - Current ROS2 support package only include one sample node for DRP-AI usage.
 - Encapsulate RZ/V2H H/W computer vision & AI accelerators, video codec, AI inference engine, and supported AI models as ROS2 nodes that are easy and quick to use in robotic s/w development.
 - Adapt more sensors including monocular camera, stereo camera (Intel RealSense, etc), RGBD camera in forms of ROS nodes.
 - [NVIDIA Issac ROS](#) and [D-Robotics](#) (spun off from Horizon Robotics) are good references.
 - Explore more for robotics application based on the above framework.
- ❖ **Investigate and bring Micro-ROS to RT core (CR8)**
 - Micro-ROS extends the ROS2 framework to MCU, allowing seamless integration with other ROS2 nodes via Micro-ROS agent (ROS2 node on app core).
 - Officially cooperates with Renesas and supports RA6M5.
 - Need investigation and porting to the RZ/V2H RT core.
 - Provide sample micro-ROS client nodes for processing sensor data (e.g., IMU) or managing motion control.



RDK ROS2 PACKAGES OVERVIEW

- Based on Ubuntu 24.04 + ROS2 Jazzy

Provide AI model abstractions and implementation for RZ/V2H DRP-AI

Perform object detection with various detection models

A collection of ROS launch files for demonstrating and experimenting with RZ/V2H platform capabilities

```
baoandu@us-srv:~/ws/rz_ros2_ws$ tree -L 2
.
├── src
│   ├── arm_hand_control
│   ├── foxglove_keypoint_publisher
│   ├── gitlab-profile
│   ├── inspire_rh56_urdf
│   ├── rzv_model
│   ├── rzv_object_detection
│   ├── rzv_playground
│   └── rzv_pose_estimation
└──
```

Control robotic hands via gesture commands and landmark tracking

Publish key points (landmarks or bounding boxes) as Foxglove image annotations for visualization

URDF models for INSPIRE RH56 robotic hand for sim and visualization

Perform pose estimation (e.g., hand landmark)

LINUX DISTRO

Linux is central to the robotics S/W stack, powered by open-source middleware, frameworks, and libraries that drive innovation and flexibility.

❖ Update Yocto (toolset to build Linux distro) ★★★

- To integrate the latest Robotics middleware, ROS2 Humble or later (via the meta-ros2 layer), along with a new kernel, Yocto 5.0 (Scarthgap) is required.
- The new release may also include a new kernel (6.x), which is in high demand among non-automotive customers. (Currently, Kernel 5.10 is in use.)
- R-Car Gen4 are still using Yocto 3.1 (Dunfell). Official update schedule is currently unclear.
- R-Car Gen5 will support Yocto 5.0.

❖ Integrate ROS2 (M/W framework for robotic systems development) ★★★

- A must-have for the robotics S/W stack
- R-Car Gen4 doesn't include it.
- R-Car Gen5 includes it in the SDK since v4.14.0 (but for now only for the simulator env)
- R-Car Gen5 will include it in the QNX SDK as well.

❖ Integrate GStreamer (framework for media processing pipeline) ★★

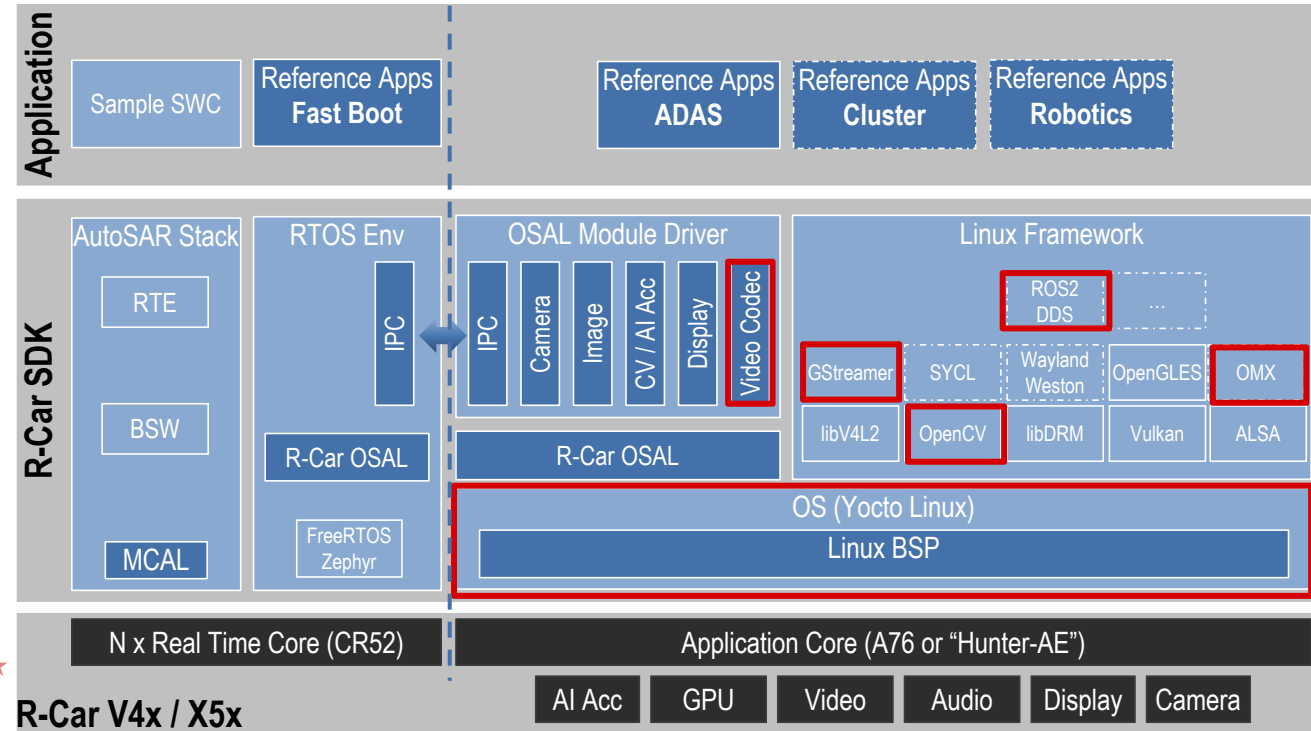
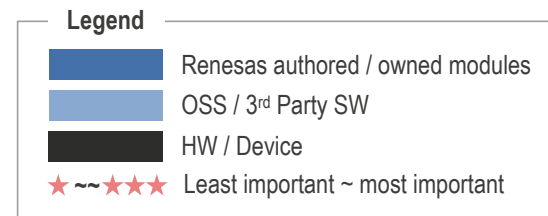
- Used in some ROS2 nodes to handle multimedia data
- R-Car Gen4 doesn't include it, but it is easy to add by simply reconfiguring Yocto.

❖ Integrate H/W video codec to standard framework as OpenOMX, Gstreamer ★ or FFmpeg

- Essential for encoding and decoding video streams within robotic applications
- The codec is currently offered as proprietary module and API.

❖ Accelerate OpenCV by H/W ★

- Utilized to process visual info and facilitate computer vision tasks in robotic applications.
- Included in R-Car Gen4, however no consideration for H/W accelerator like RZ/V2H DRP or NVIDIA CUDA, despite its potential H/W capabilities for acceleration.



WS077 DEXTEROUS HAND SOLUTION

System Overview

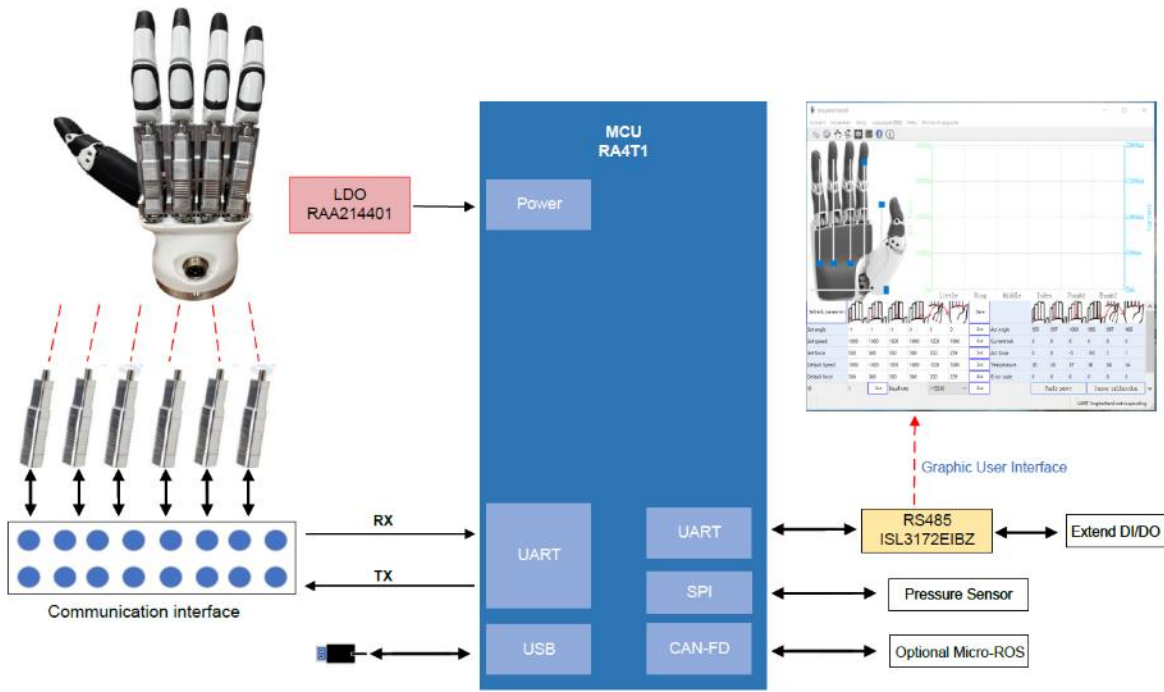
This robotic hand control system is designed for precise and dexterous movements, with 6 cylinders controlling the 5 fingers and additional actuators for flexibility. It features a user-friendly GUI for easy operation and real-time feedback, even for first-time users. With multiple communication interfaces, it supports seamless connectivity and sensor integration for advanced functions, making it versatile for prosthetics, automation, and robotics research.

System Benefits

- The system supports multiple communication protocols, including CAN-FD, which ensures reliable, high-speed data transfer. It integrates micro-ROS, bringing the advanced features of ROS 2 to the microcontroller level.
- The 6-cylinder control mechanism ensures precise and independent movement of each finger, enabling dexterous hand functions for various applications.
- With modular design and multiple sensor integration capabilities, the system can be easily expanded to support advanced features like adaptive grip, tactile sensing, and environmental interaction.

Target Application

- Human-Robot(Dexterous hand)
- Manufacturing automation



Device Category	P/N	Key Features
MCU	RA4T1	Renesas 100MHz Arm Cortex-M33 with TrustZone
Power	RAA214401	150mA Wide Input Range Ultra-Low Quiescent Current Linear Regulator
Analog	ISL3172EIBZ	±15kV IEC61000 ESD protected, 3.3V powered single transceivers

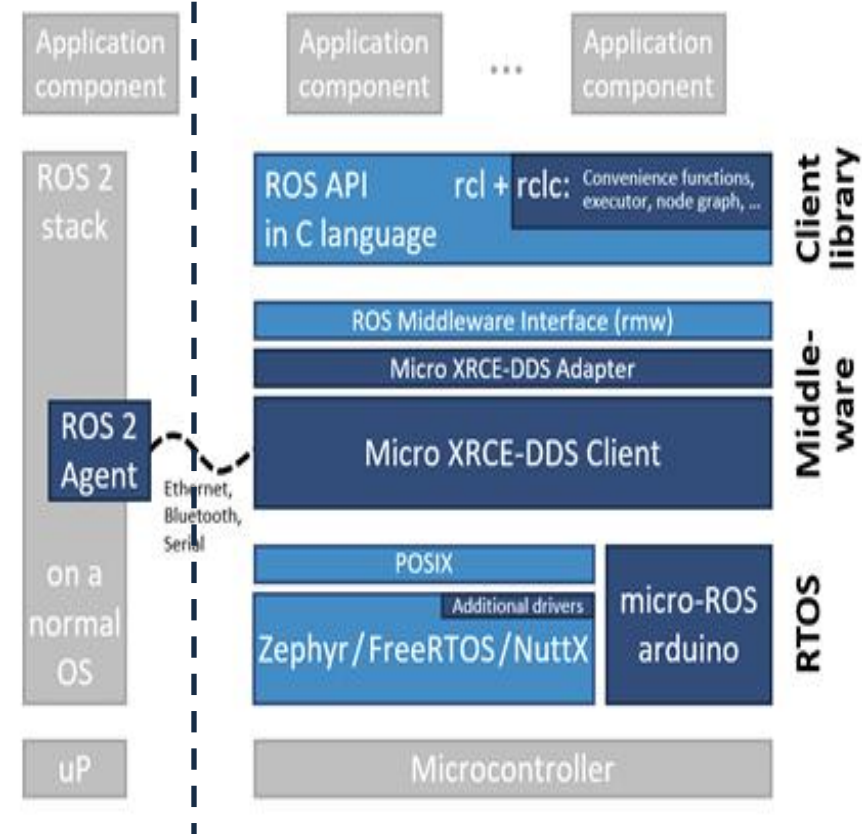
DEXTEROUS HAND SOLUTION SW ARCHTECT PLAN



MICRO-ROS STACK

Humanoid Robotic Middleware initiative

Agent



Micro-ROS is a robotic operating system (ROS) framework extension specifically designed for microcontrollers, bringing the capabilities of ROS2 to resource-constrained embedded devices.

- **Lightweight design for resource-constrained devices**
Micro-ROS is optimized for resource-constrained microcontrollers, enabling it to run on low-power, low-memory devices
- **Seamless integration with ROS2**
Fully compatible with ROS2 communication protocols, allowing microcontroller devices to participate directly in distributed robotic systems.
- **Real-time support**
Provides real-time capabilities based on RTOS, suitable for latency-sensitive scenarios
- **Modularity and Cross-Platform Support**
Supports modular design and various hardware platforms and RTOS, flexibly adapting to different application needs



WS078 ROBOTIC STANDARD ARM SYSTEM

System Overview

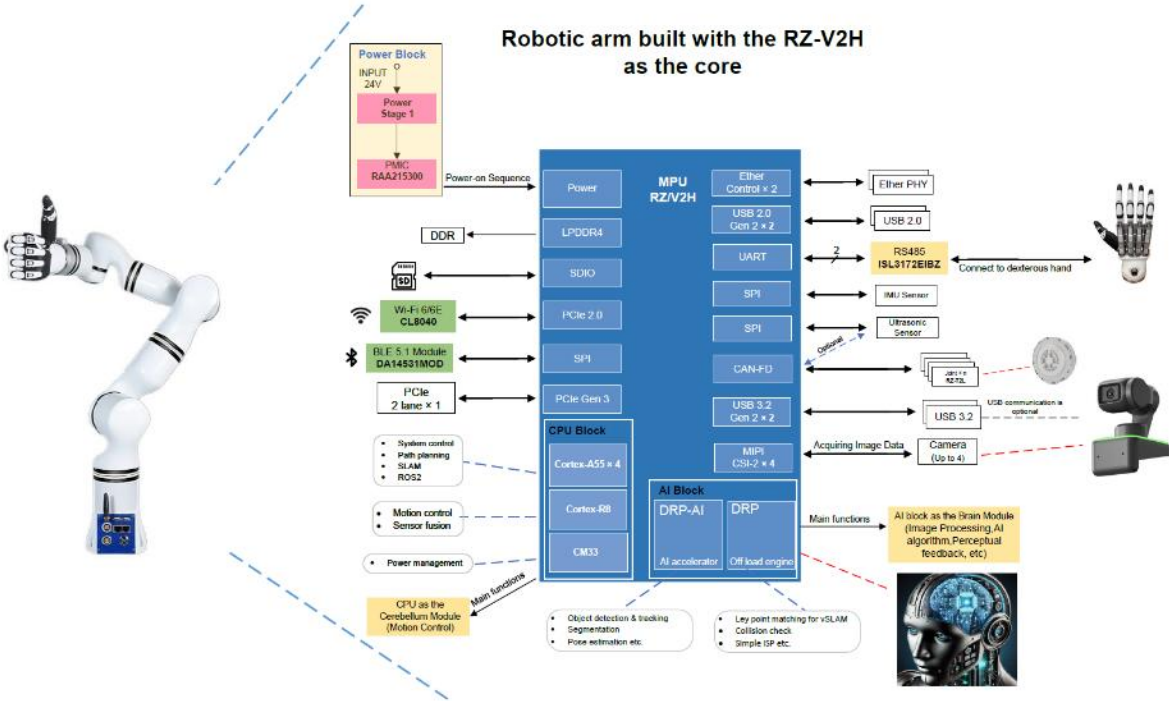
This robotic arm system utilizes a collaborative "brain and cerebellum" architecture powered by the RZ/V2H processor. The AI module (brain) with DRP-AI accelerators handles perception and decision-making, while the CPU module (cerebellum) ensures real-time motion control, sensor fusion, and path planning. Additionally, it supports multiple communication protocols, enabling seamless integration with other humanoid robot modules, such as joints and dexterous hands, forming a versatile and complete system.

System Benefits

- Powered by DRP-AI accelerators, the system achieves real-time object detection, segmentation, and pose estimation. These features provide the robotic arm with robust perception and decision-making abilities, crucial for tasks requiring high precision and adaptability.
- With support for protocols like CAN-FD, Ethernet, SPI, and RS485, the system seamlessly integrates with other modules, such as joints and dexterous hands, forming a unified and functional humanoid robot system. This flexibility makes it adaptable for different applications and easy to expand.
- The multi-core Cortex-A55 and Cortex-R8 processors enable real-time motion control and sensor fusion, ensuring smooth and responsive robotic operations.

Target Application

- Prosthetic Control System
- Human-Robot
- Dangerous Environment Maintenance



Device Category	P/N	Key Features
SoC	RZ/V2H R9A09G057H48GBG#AC0	Vision AI MPU With Renesas DRP-AI High-efficiency AI Accelerator and OpenCV Image Processing Accelerator Enabling Real-time Object Recognition
Power	RAA215300	High Performance 9-Channel PMIC
Connectivity	CL8040	Wi-Fi 6 Concurrent Dual Band 4T4R PCIe Chip
	DA14531MOD-00F0100C	SmartBond TINY™ Bluetooth® Low Energy Module

WS079 R-CAR V4H ROBOTIC ARM HIGH COMPUTING SYSTEM

System Overview

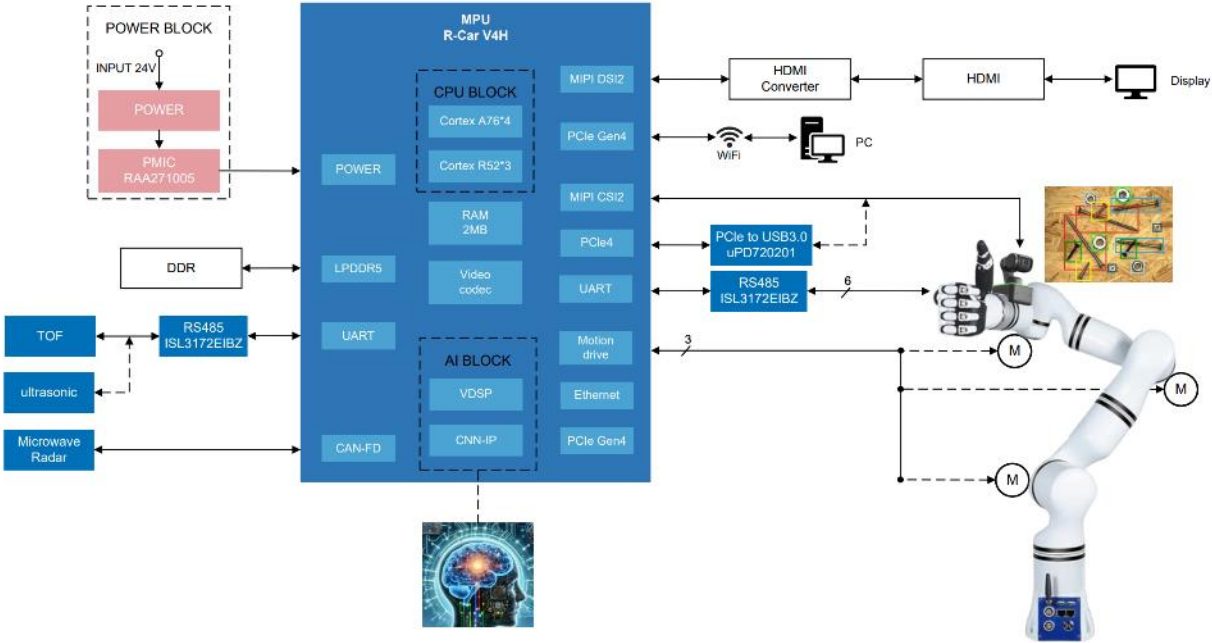
The R-Car V4H offers up to 34 TOPS of highly efficient deep learning performance. It enables high-speed image recognition and processing. The system supports real-time object detection and precise robot motion control, making it suitable for next-generation industrial automation and AI-driven robotics. It integrates a rich set of sensors, such as TOF, ultrasonic and microwave radar to provide environmental awareness and situational understanding.

System benefits

- Up to 34 TOPS of highly efficient deep learning performance. It uses ultra-low power consumption and requires only passive air-cooling, both of which help reduce the total bill of materials (BOM).
- Dual-core real-time processors ensure precise motion control and rapid response to sensor data.
- Integrates TOF, ultrasonic and microwave radar sensors for enhanced situational awareness and environmental mapping.

Target Applications

- Industrial Robot
- Healthcare Robot
- AGV
- Smart Agriculture



Device Category	P/N	Key Features
SoC	R-CAR V4H R8A779G3LA01BA	Vision AI MPU With Renesas DRP-AI High-efficiency AI Accelerator and OpenCV Image Processing Accelerator Enabling Real-time Object Recognition
Power	RAA271005	Provides multiple buck and low dropout regulators (LDOs) to power various system components such as processors, memory and peripherals, operating from 4.5V to 42V.
Analog	ISL3172EIBZ	UART Low Energy Module
	uPD720201	Pcie4.0 to USB3.0 Module

WS080 HUMANOID ROBOT SYSTEM

System Overview

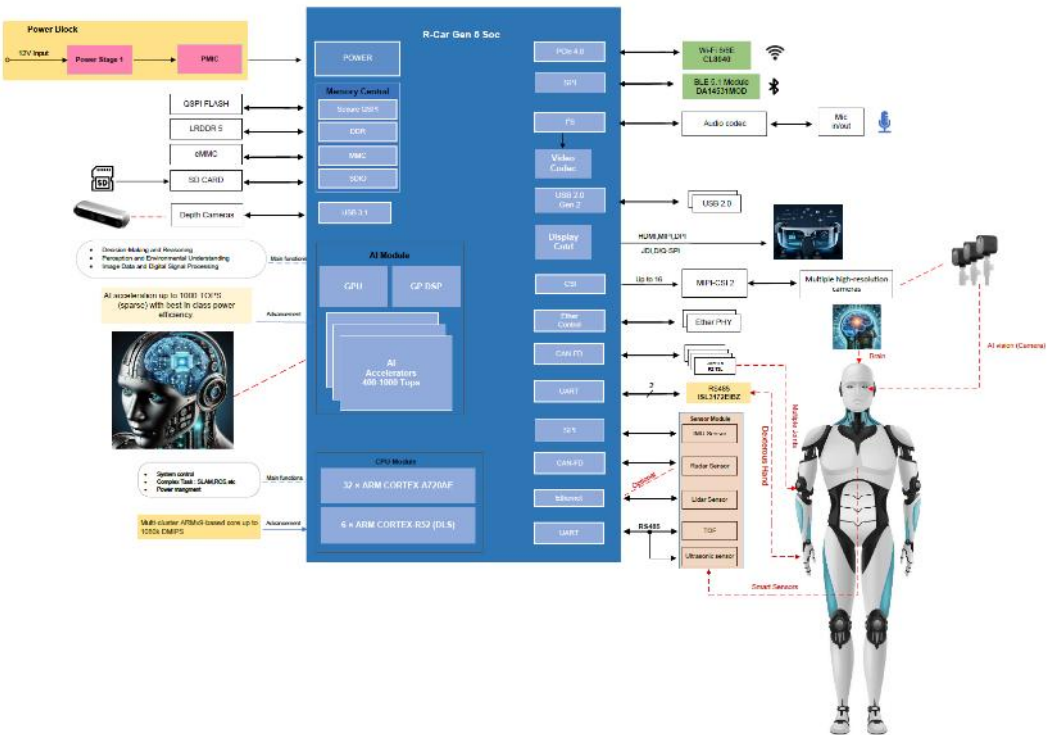
This humanoid robot system leverages Renesas' most advanced R-Car Gen 5 SoC, utilizing its 3nm process technology for an optimal balance of ultra-high AI performance (up to 1000 TOPS) and low power consumption. The system integrates a variety of high-speed communication interfaces to ensure seamless, real-time coordination between critical components, such as the robot's joints, cerebellum, and dexterous hands, enabling smooth, precise, and human-like movements.

System Benefits

- The system uses the R-Car Gen 5 SoC with AI accelerators delivering up to 1000 TOPS. This enables real-time execution of complex AI tasks like vision processing, object recognition, and decision-making, ensuring the robot is adaptive and intelligent in dynamic environments.
- The X5H chip's CPU combines a 32-core ARM Cortex-A720AE, delivering up to 1080k DMIPS for complex AI tasks, with a 6-core ARM Cortex-R52, providing 129k DMIPS for real-time operations.
- The system boasts a diverse array of high-speed communication interfaces, including CAN-FD, RS485, SPI, UART, and Ether PHY, which are critical for ensuring low-latency, real-time communication between the robot's key components.
- Using 3nm process technology, ensuring superior energy efficiency without compromising performance. This low-power design extends the operational lifespan of battery-powered robots and reduces heat generation, making the system suitable for long-term, mobile, or field-deployed applications.

Target Applications

- Human robot
- Autonomous Navigation



Device Category	P/N	Key Features
SoC	R-Car Gen5	Renesas 100MHz Arm Cortex-M33 with TrustZone
Power	RAA214401	150mA Wide Input Range Ultra-Low Quiescent Current Linear Regulator
Analog	ISL3172EIBZ	±15kV IEC61000 ESD protected, 3.3V powered single transceivers